



# Airplane Performance

Airplane Characteristics  
Airport Analysis  
Study Flight Rules

*The statements contained herein are based on good faith assumptions and provided for general information purposes only. These statements do not constitute an offer, promise, warranty or guarantee of performance. Actual results may vary depending on certain events or conditions. This document should not be used or relied upon for any purpose other than that intended by Boeing.*

*Boeing is a trademark of The Boeing Company.*

*Copyright © 2014 Boeing. All rights reserved.*

# Objectives

- Overview of primary components of airplane performance analysis to provide:
  - How airplane characteristics affect airplane performance
  - How airport characteristics impact airplane performance
  - How study flight rules impact airplane performance

# First steps: three main components of airplane analysis (also called a performance study)

In order for a performance study to be meaningful and useful to an airline, a set of performance ground rules must be developed that will be representative of the airline's operational standards



**Airplane Characteristics**



**Airport Analysis**

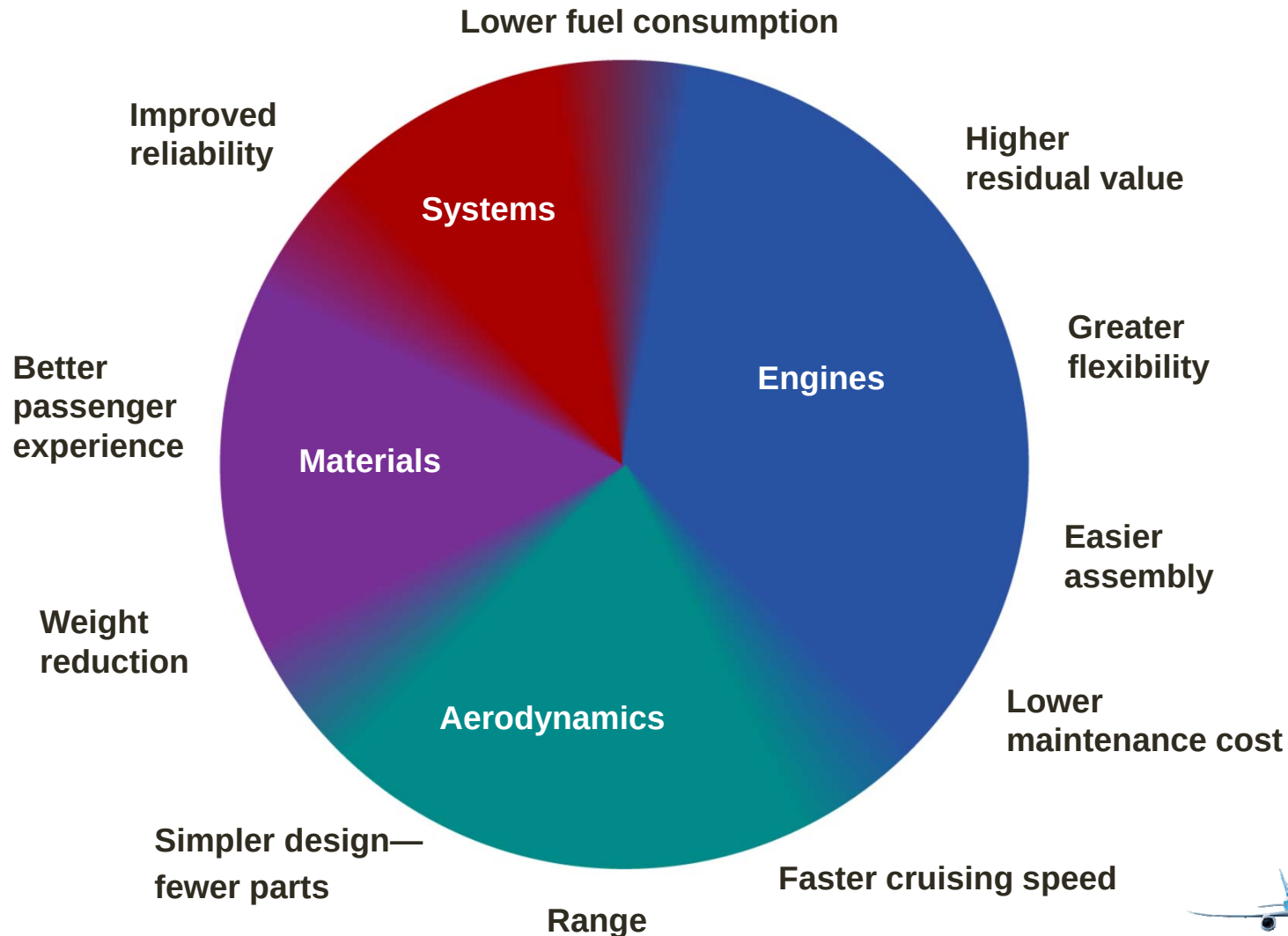


**Study Flight Rules**

Copyright © 2013 Boeing. All rights reserved.

# Advanced technology contributes to new airplane efficiencies

*Numerous benefits and new design possibilities*

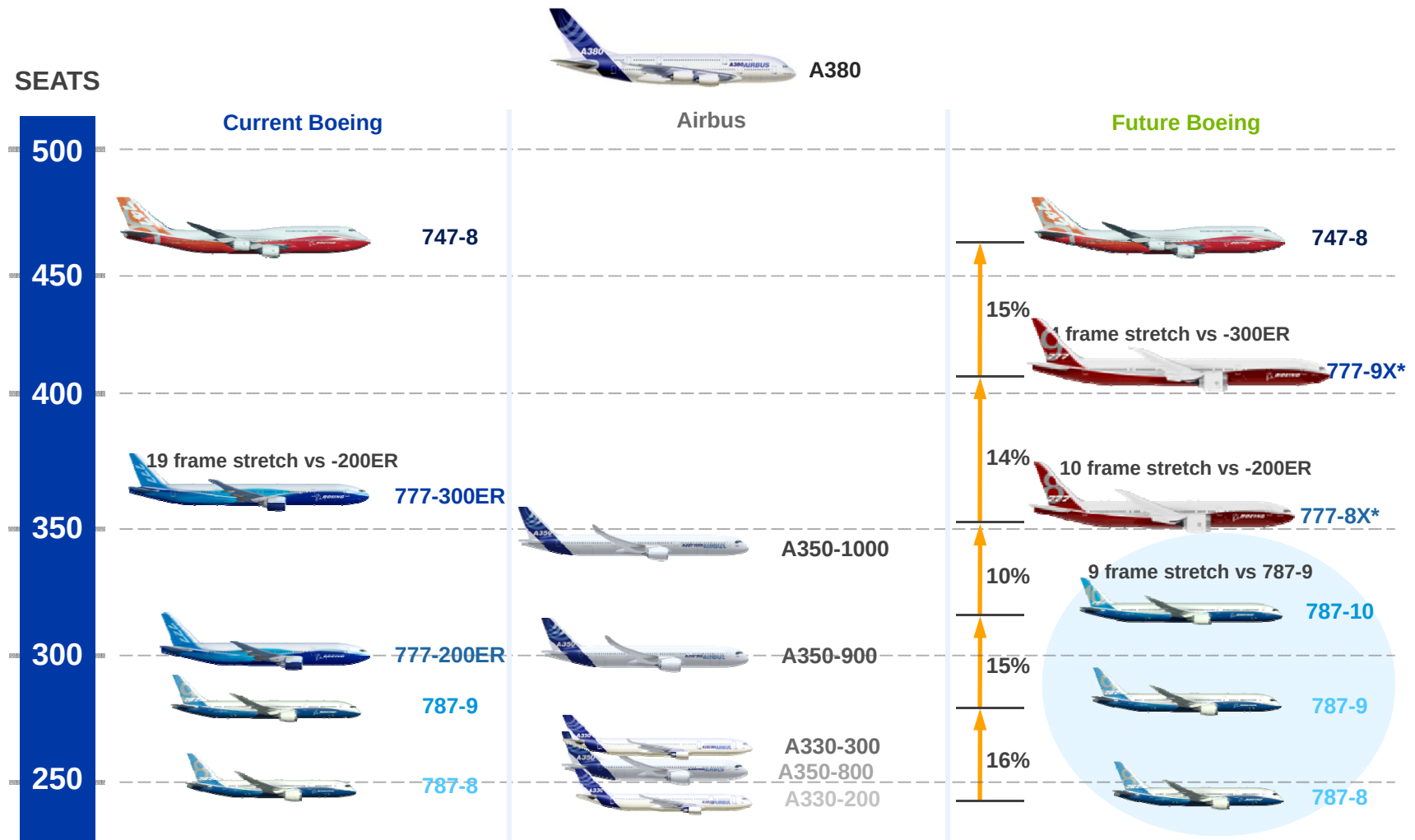




Copyright © 2013 Boeing. All rights reserved.

# Broad market coverage

The newest, most efficient, long range family



\* Product Development Study

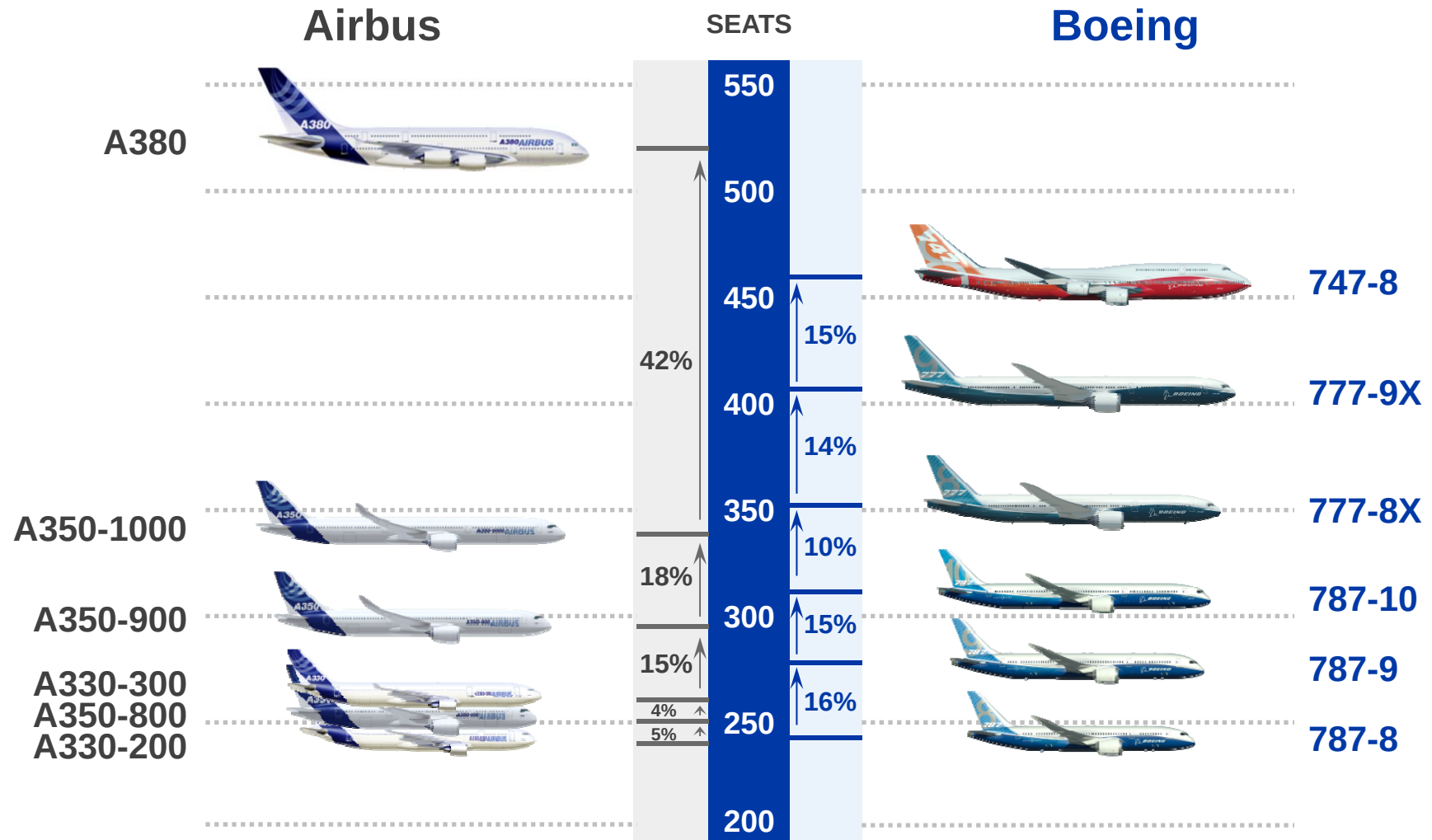
Copyright © 2014 Boeing. All rights reserved.

BOEING PROPRIETARY

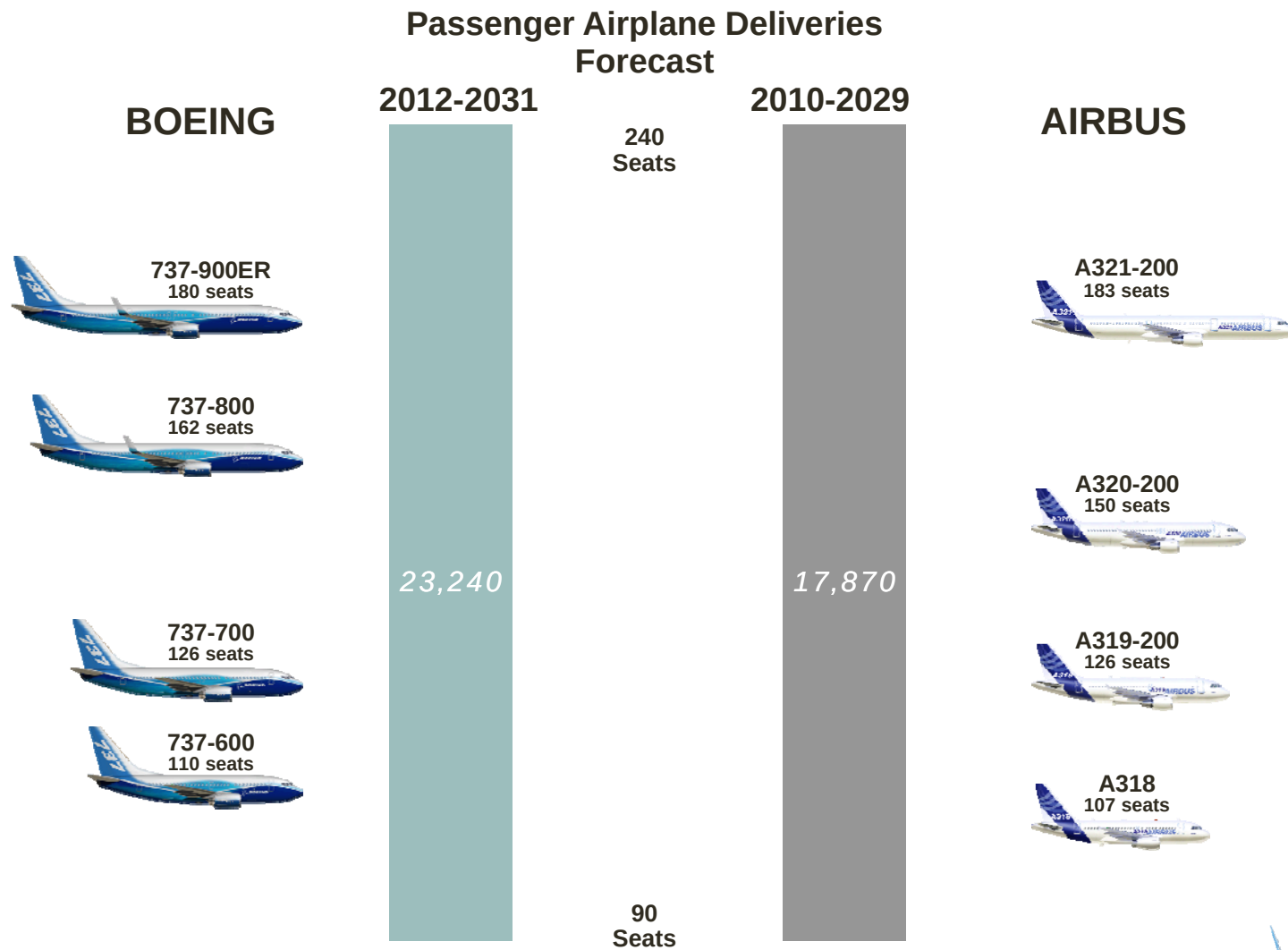
6

# Family comparison chart example

This is a subhead, usage is optional



# Single aisle airplanes





# Airplane Performance

**Airplane Characteristics**

Airport Analysis  
Study Flight Rules

# Airplane characteristics: Boeing mission analysis program route analysis

## Mission Performance Summary

|                                                                              |           |                    |                   |                        |                     |                 |
|------------------------------------------------------------------------------|-----------|--------------------|-------------------|------------------------|---------------------|-----------------|
| <b>737-700W Gross Weight Comparison</b>                                      |           |                    | Max Taxi Wt:      | 155,000 lb             | Structural Payload: | 36,610 lb       |
| <b>Typical Intl Rules; 2% Fuel Markup</b>                                    |           |                    | Max TO Wt:        | 154,500 lb             | Study Payload:      | 36,610 lb       |
| <b>Mission Data</b>                                                          |           |                    | Max Land Wt:      | 129,200 lb             | Passenger Seats:    | 126 @ 210 lb ea |
| Enroute Wind Reliability: 85%                                                | Airframe: | <b>737-700W</b>    | Max Zero Fuel Wt: | 121,700 lb             | Pax Bags Payload:   | 26,460 lb       |
| Enroute Wind & Temps Season: Annual                                          | Engine:   | <b>CFM56-7B22E</b> | Op Empty Wt:      | 85,090 lb              | Rev Cargo:          | 10,150 lb       |
|                                                                              | Interior: | <b>126</b>         | Fuel Capacity:    | 6,875 gal              |                     |                 |
|                                                                              |           |                    | Fuel Wt:          | 46,063 lb @ 6.7 lb/gal |                     |                 |
| APS job.APNav / 737-700W / CFM56-7B22E / D6388275REVA / 8/30/2011 1:00:56 PM |           |                    |                   |                        |                     |                 |

| Track      | Rwy | Dist<br>(nm) | ESAD<br>(nm) | Cruise<br>Altitude<br>(100ft) | Cruise<br>Wind<br>(kt) | Cruise<br>Delta<br>Temp<br>(°C) | Mach | Block<br>Time<br>(h) | Takeoff<br>Weight<br>(lb) | Landing<br>Weight<br>(lb) | Block<br>Fuel<br>(lb) | Reserve<br>Fuel<br>(lb) | PAX | Cargo<br>(lb) | Payload<br>(lb) | Altn | Altn<br>Dist<br>(nm) |
|------------|-----|--------------|--------------|-------------------------------|------------------------|---------------------------------|------|----------------------|---------------------------|---------------------------|-----------------------|-------------------------|-----|---------------|-----------------|------|----------------------|
| DEN to JFK | 34L | 1,436        | 1,363        | 390                           | 25                     | 5                               | LRC  | 3.450                | 141,308                   | 125,510                   | 16,155                | 3,810                   | 126 | 10,150        | 36,610          | EWR  | 18                   |
| JFK to DEN | 13R | 1,436        | 1,724        | 380/400                       | -78/-76                | 6/5                             | LRC  | 4.227                | 146,748                   | 126,361                   | 20,743                | 4,660                   | 126 | 10,151        | 36,610          | COS  | 63                   |
| SEA to LAX | 34L | 846          | 885          | 390                           | -21                    | 5                               | LRC  | 2.404                | 138,897                   | 127,934                   | 11,320                | 6,230                   | 126 | 10,154        | 36,610          | LAS  | 205                  |
| LAX to SEA | 25R | 846          | 920          | 400                           | -38                    | 5                               | LRC  | 2.475                | 138,179                   | 126,823                   | 11,714                | 5,124                   | 126 | 10,149        | 36,610          | PDX  | 112                  |
| MDW to MCO | 31C | 878          | 896          | 390                           | -10                    | 4                               | LRC  | 2.434                | 137,120                   | 126,172                   | 11,305                | 4,471                   | 126 | 10,151        | 36,610          | TPA  | 70                   |
| MCO to MDW | 17L | 878          | 976          | 400                           | -48                    | 3                               | LRC  | 2.603                | 137,031                   | 125,213                   | 12,175                | 3,515                   | 126 | 10,148        | 36,610          | ORD  | 14                   |
| MDW to PHX | 31C | 1,277        | 1,500        | 380/400                       | -70/-68                | 5/4                             | LRC  | 3.759                | 142,453                   | 124,898                   | 17,913                | 5,109                   | 126 | 8,238         | 34,698 ATOW     | TUS  | 96                   |
| PHX to MDW | 26  | 1,277        | 1,232        | 390                           | 17                     | 5                               | LRC  | 3.168                | 139,902                   | 125,351                   | 14,908                | 3,652                   | 126 | 10,149        | 36,610          | ORD  | 14                   |

# Airplane characteristics

- Configuration
- Gross Weights
- Payload
  - Interior arrangement
  - Engine type and thrust
  - Design weight limit and empty weight
- Thrust Requirements

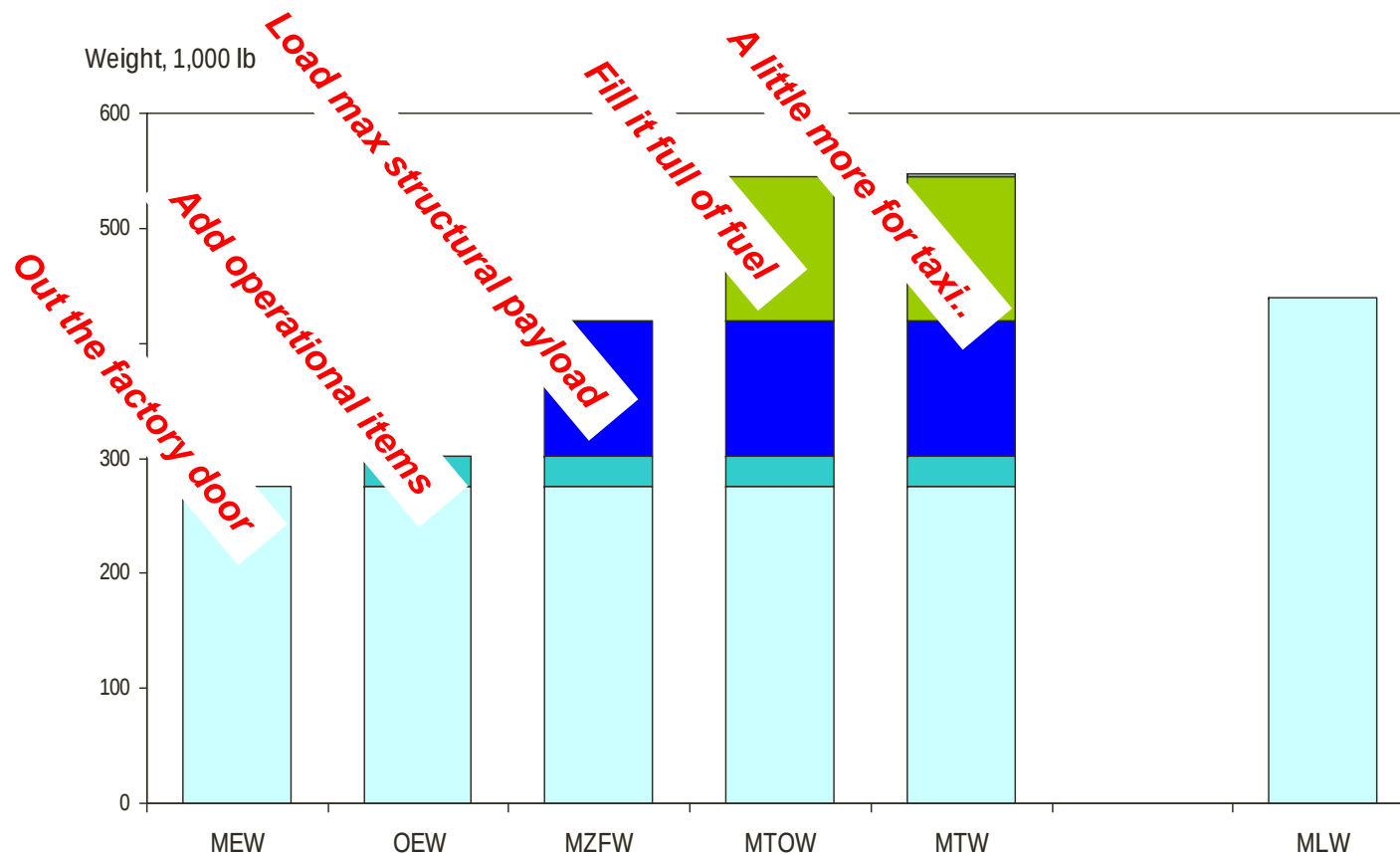


# Airplane characteristics: 777-200

Example for 305 passengers (24 first class, 54 business class, and 227 tourist class)

|                                                                              |                     |   | Weight (lb) |
|------------------------------------------------------------------------------|---------------------|---|-------------|
| <b>Baseline Manufacturer's Empty Weight (MEW)</b>                            |                     |   | 273,500     |
| Configuration specification, D019W005, Rev A (Date, TBD)                     |                     |   |             |
| 305 (24 F/54 B/227 Y)                                                        | Interior            |   |             |
| Rolls-Royce Trent 884                                                        | Engines             |   |             |
| 508,000 lb (230,424 kg)                                                      | Maximum taxi weight |   |             |
| 31,000 USG (117,347 L)                                                       | Fuel capacity       |   |             |
| <b>Customer Changes:</b>                                                     |                     |   | 1,641       |
| Interior change to 305 passengers (24 FC/54 BC/227 TC) (Ref: LOPS ICX-2776D) |                     | 0 |             |
| Customer options allowance                                                   |                     |   | 1,641       |
| <b>Manufacturer's Empty Weight (MEW)</b>                                     |                     |   | 275,141     |
| Standard items allowance                                                     |                     |   | 5,765       |
| Unusable fuel                                                                | 475                 |   |             |
| Oil                                                                          | 175                 |   |             |
| Oxygen equipment                                                             | 91                  |   |             |
| Miscellaneous equipment                                                      | 71                  |   |             |
| Galley structure and fixed inserts                                           | 1,953               |   |             |
| Operational items allowance                                                  |                     |   | 21,159      |
| Crew and crew baggage                                                        | 1,760               |   |             |
| Flight crew (2)                                                              | 340                 |   |             |
| Cabin crew (8)                                                               | 1,120               |   |             |
| Baggage (10)                                                                 | 250                 |   |             |
| Pilot briefcases (2)                                                         | 50                  |   |             |
| Catering allowance                                                           | 10,010              |   |             |
| First class (24 at 110 lb each)                                              | 2,640               |   |             |
| Business class (54 at 44 lb each)                                            | 2,376               |   |             |
| Tourist class (227 at 22 lb each)                                            | 4,994               |   |             |
| Passenger service equipment (305)                                            | 915                 |   |             |
| Potable water, U.S. gallons (218)                                            | 1,816               |   |             |
| Waste tank disinfectant                                                      | 150                 |   |             |
| Emergency equipment (including overwater equipment)                          | 1,968               |   |             |
| Cargo system                                                                 | 4,540               |   |             |
| Forward 96- x 125-pallets (6)                                                | 1,740               |   |             |
| Aft LD-3 containers (14)                                                     | 2,800               |   |             |
| <b>Standard and Operational Items</b>                                        |                     |   | 26,924      |
| Roundoff                                                                     |                     |   | 35          |
| <b>Operational Empty Weight (OEW)</b>                                        |                     |   | 302,100     |

# Airplane characteristics: relative relationships of design weights – typical 777-200 example



**MEW:** Maximum Empty Weight  
**OEW:** Operational Empty Weight  
**MZFW:** Maximum Zero Fuel Weight  
**MTOW:** Maximum TakeOff Weight  
**MTW:** Maximum Taxi Weight  
**MLW:** Maximum Landing Weight

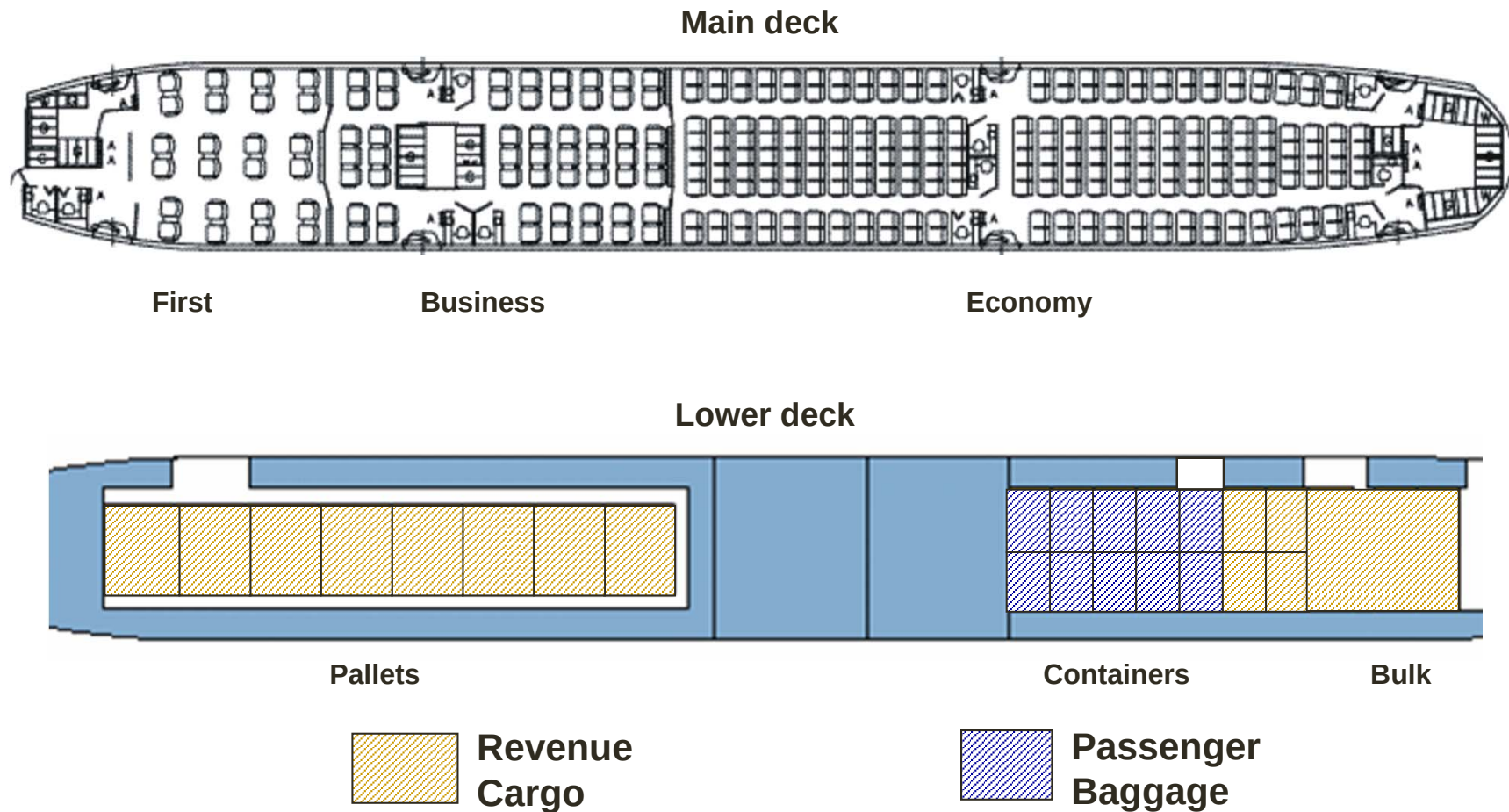
# Airplane characteristics: payload

**What are some of the characteristics of your airplanes that impact their Payload?**

- **Aircraft Configuration**
  - Interior arrangement
  - Engine type and thrust
  - Design weight limits and empty weight
- **Payload Limit Definition**



# Airplane characteristics: volume limit payload - 777-200 example with 305 seats



# Airplane characteristics: volume limit payload calculation - 777-200 example

|                                                                                             |                   |
|---------------------------------------------------------------------------------------------|-------------------|
| <b>Weight of passengers and their baggage<br/>(10 LD-3 containers required for baggage)</b> | <b>64,050 lb</b>  |
| <b>Cargo loaded in remaining LD-3 containers</b>                                            | <b>6,320 lb</b>   |
| <b>Cargo loaded on pallets</b>                                                              | <b>24,900 lb</b>  |
| <b>Cargo loaded in bulk area</b>                                                            | <b>4,800 lb</b>   |
| <b>Volume limit payload</b>                                                                 | <b>100,070 lb</b> |
| <b>Maximum structural limit payload</b>                                                     | <b>117,900 lb</b> |

# Airplane characteristics: structural payload calculation - 777-200 example with 305 seats



**Maximum Zero Fuel Weight = 420,000 lb**

**Minus**

**Operating Empty Weight = 302,100 lb**

**Equals**

**Maximum Structural Payload = 117,900 lb**

# Airplane characteristics: thrust example of available engine options supporting 777 family

Pratt & Whitney



General Electric



Rolls-Royce



|           | Takeoff thrust, lb |          | Takeoff thrust, lb |            | Takeoff thrust, lb |             |
|-----------|--------------------|----------|--------------------|------------|--------------------|-------------|
| 777-200   | 74,400             | (PW4074) | 77,000             | (GE90-77B) | 73,400             | (Trent 875) |
|           | 77,000             | (PW4077) |                    |            | 76,000             | (Trent 877) |
| 777-200ER | 84,400             | (PW4084) | 84,700             | (GE90-85B) | 83,600             | (Trent 884) |
|           | 90,000             | (PW4090) | 90,000             | (GE90-90B) | 90,000             | (Trent 892) |
|           | 97,900             | (PW4098) | 93,700             | (GE90-94B) | 93,400             | (Trent 895) |
| 777-300   | 90,000             | (PW4090) |                    |            | 83,600             | (Trent 884) |
|           | 98,000             | (PW4098) |                    |            | 90,000             | (Trent 892) |

All thrusts are Boeing equivalent.



# Airplane Performance

Airplane Characteristics

**Airport Analysis**

Study Flight Rules

# Airport analysis

## 737-700W/CFM56-7B22E HGW Takeoff Weight Capability

### 737-700W Gross Weight Comparison

- 737-700W/CFM56-7B22E
- Certification: FAA-SCAP
- MTOW: 154,500 lb
- Surface Temperature Season: Annual
- Surface Temperature Reliability: ADM
- C.G.: 15 %MAC Alternate
- A/C: OFF
- Optimum Performance
- Runway Contamination: DRY

| Airport <sup>1</sup><br>(CODE) | Runway | Pressure<br>Altitude<br>(ft) | Field<br>Length<br>(ft) | Slope<br>(%) | Clearway<br>(ft) | Stopway<br>(ft) | Obstacle<br>Height <sup>2</sup><br>(ft) | Obstacle<br>Distance <sup>2</sup><br>(ft) | Temperature<br>(°C) | Takeoff<br>Weight <sup>3</sup><br>(lb) |
|--------------------------------|--------|------------------------------|-------------------------|--------------|------------------|-----------------|-----------------------------------------|-------------------------------------------|---------------------|----------------------------------------|
| Chicago<br>(MDW)               | 31C    | 612                          | 6,522                   | -0.08        | 0                | 0               | 17                                      | 182                                       | 14.4 †              | 142,453 (o2)                           |
|                                |        |                              |                         |              |                  |                 | 37                                      | 573                                       |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 59                                      | 1,880                                     |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 139                                     | 6,743                                     |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 200                                     | 9,106                                     |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 283                                     | 56,486                                    |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 293                                     | 59,384                                    |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 298                                     | 69,917                                    |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 501                                     | 71,001                                    |                     |                                        |
| Denver<br>(DEN)                | 34L    | 5,324                        | 16,000                  | -0.03        | 0                | 0               |                                         |                                           | 17.9 †              | 154,500 (s)                            |
| Los Angeles<br>(LAX)           | 25R    | 94                           | 12,091                  | 0.22         | 0                | 0               | 16                                      | 908                                       | 21.2 †              | 154,500 (s)                            |
|                                |        |                              |                         |              |                  |                 | 84                                      | 3,907                                     |                     |                                        |
| New York<br>(JFK)              | 13R    | 13                           | 14,572                  | 0            | 0                | 0               | 12                                      | 98                                        | 16.1 †              | 154,500 (s)                            |
|                                |        |                              |                         |              |                  |                 | 87                                      | 4,954                                     |                     |                                        |
| Orlando<br>(MCO)               | 17L    | 90                           | 9,000                   | 0            | 0                | 0               | 490                                     | 88,872                                    | 28.3 †              | 154,500 (s)                            |
| Phoenix<br>(PHX)               | 26     | 1,135                        | 11,489                  | -0.21        | 0                | 0               | 14                                      | 63                                        | 29.8 †              | 154,500 (s)                            |
|                                |        |                              |                         |              |                  |                 | 16                                      | 82                                        |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 22                                      | 118                                       |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 50                                      | 2,163                                     |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 60                                      | 2,302                                     |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 66                                      | 3,020                                     |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 219                                     | 10,944                                    |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 424                                     | 24,553                                    |                     |                                        |
| Seattle<br>(SEA)               | 34L    | 356                          | 8,500                   | 0.69         | 0                | 0               | 113                                     | 5,995                                     | 15.1 †              | 154,500 (s)                            |
|                                |        |                              |                         |              |                  |                 | 266                                     | 53,612                                    |                     |                                        |
|                                |        |                              |                         |              |                  |                 | 634                                     | 55,253                                    |                     |                                        |

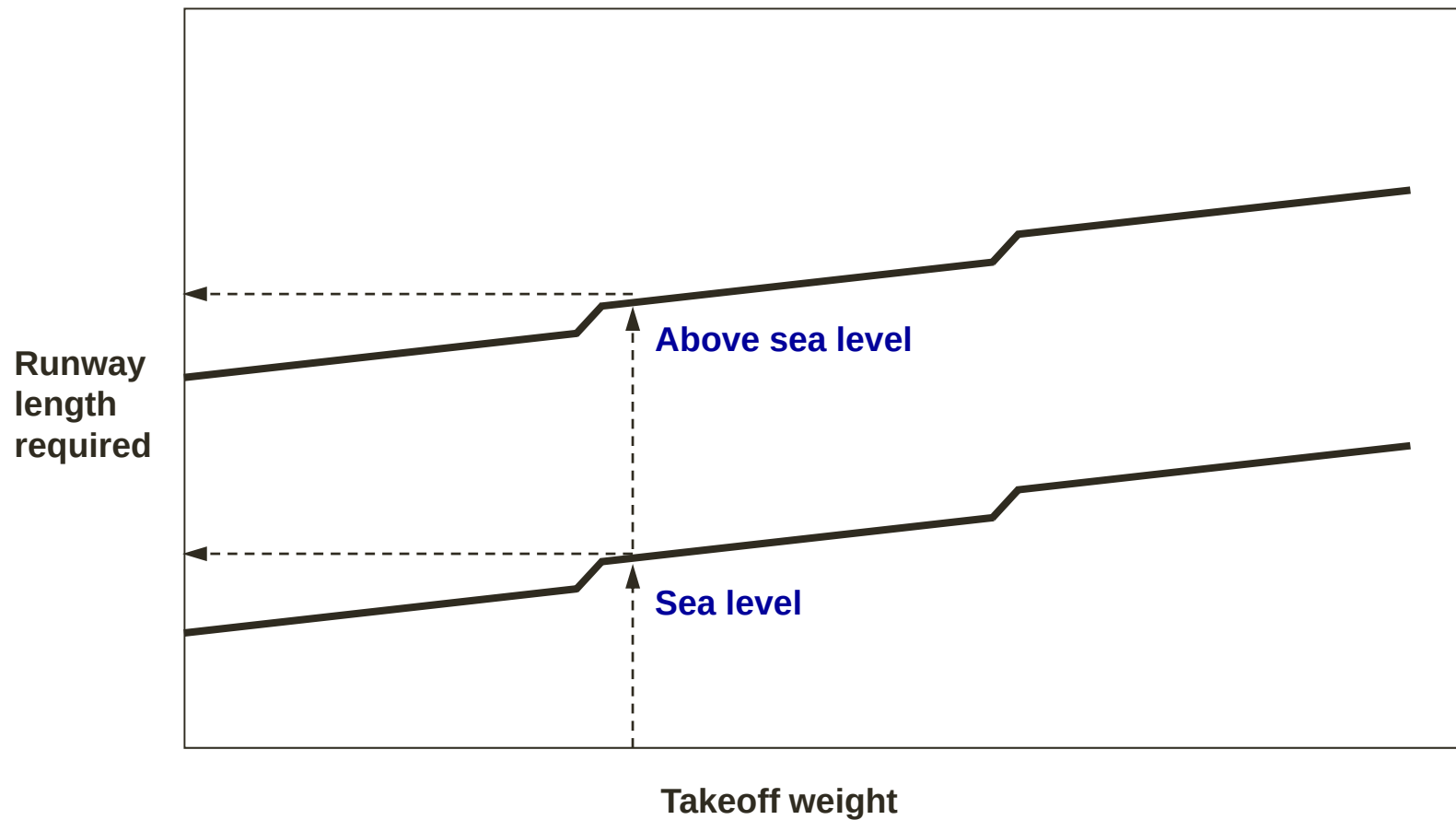
# Airport analysis: takeoff performance

Takeoff weights are set by the most restrictive of

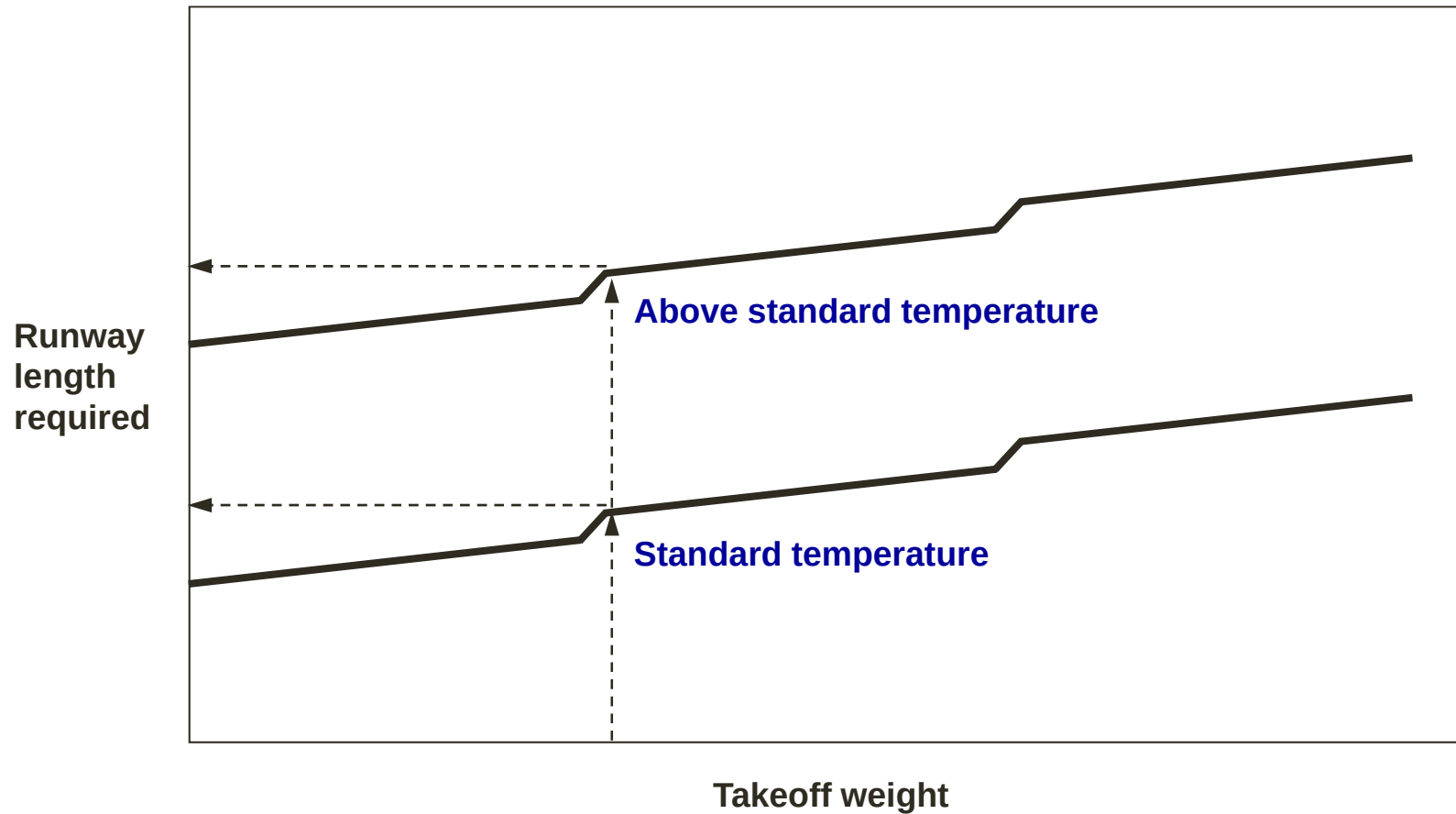
- Maximum certified takeoff weight
- Field length limiting weight
- Second segment limiting weight
- Minimum control speed
- Tire speed limiting weight
- Brake energy limiting weight
- Obstacle clearance limiting weight



# Airport analysis: takeoff field length at specified temperature



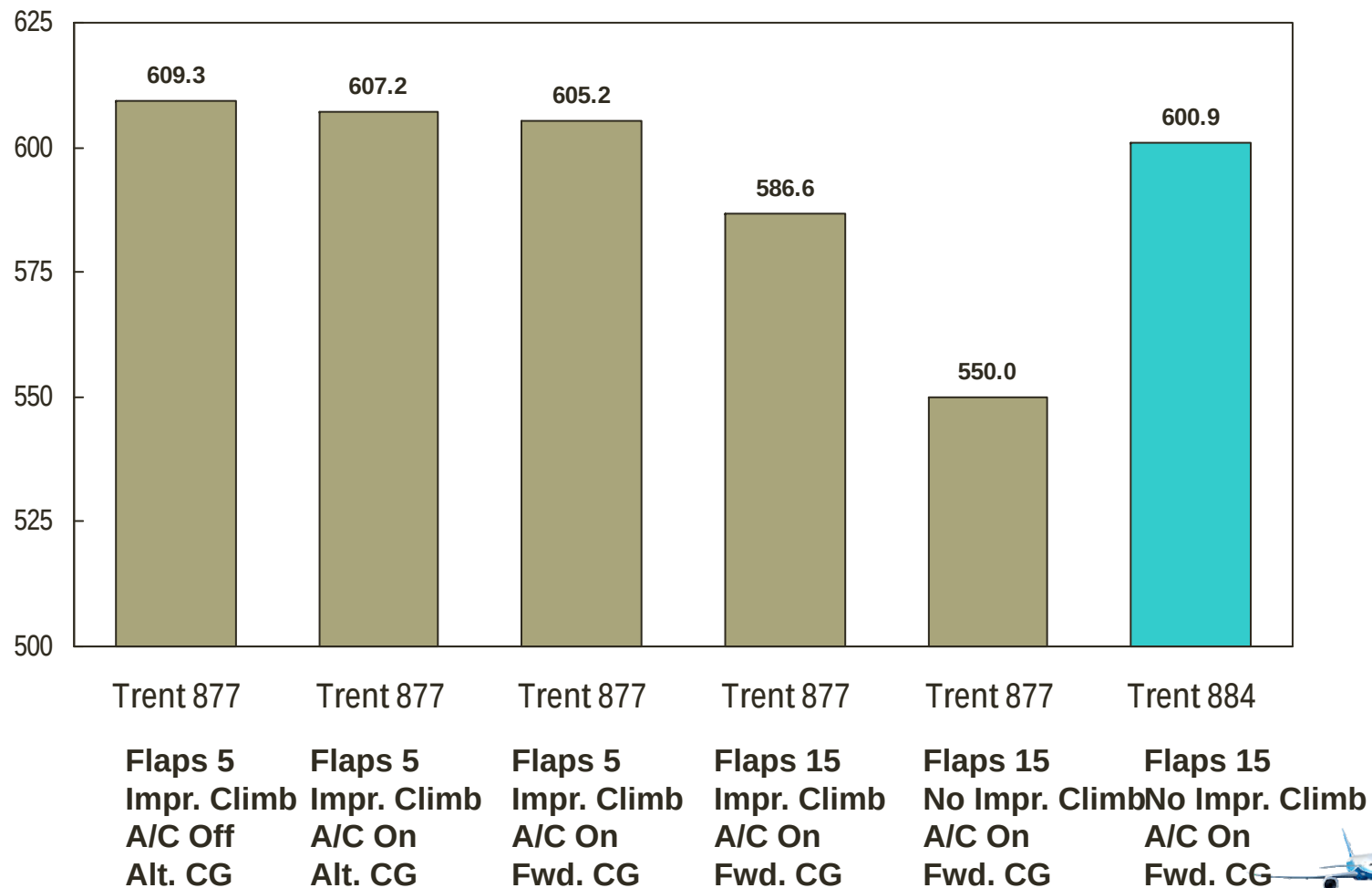
# Airport analysis: takeoff field length at constant elevation



# Airport analysis: 777-200ER takeoff weight capability example

Dubai Runway 12L: Winter ADM Temp = 23.7° C

Takeoff weight, 1,000 lb





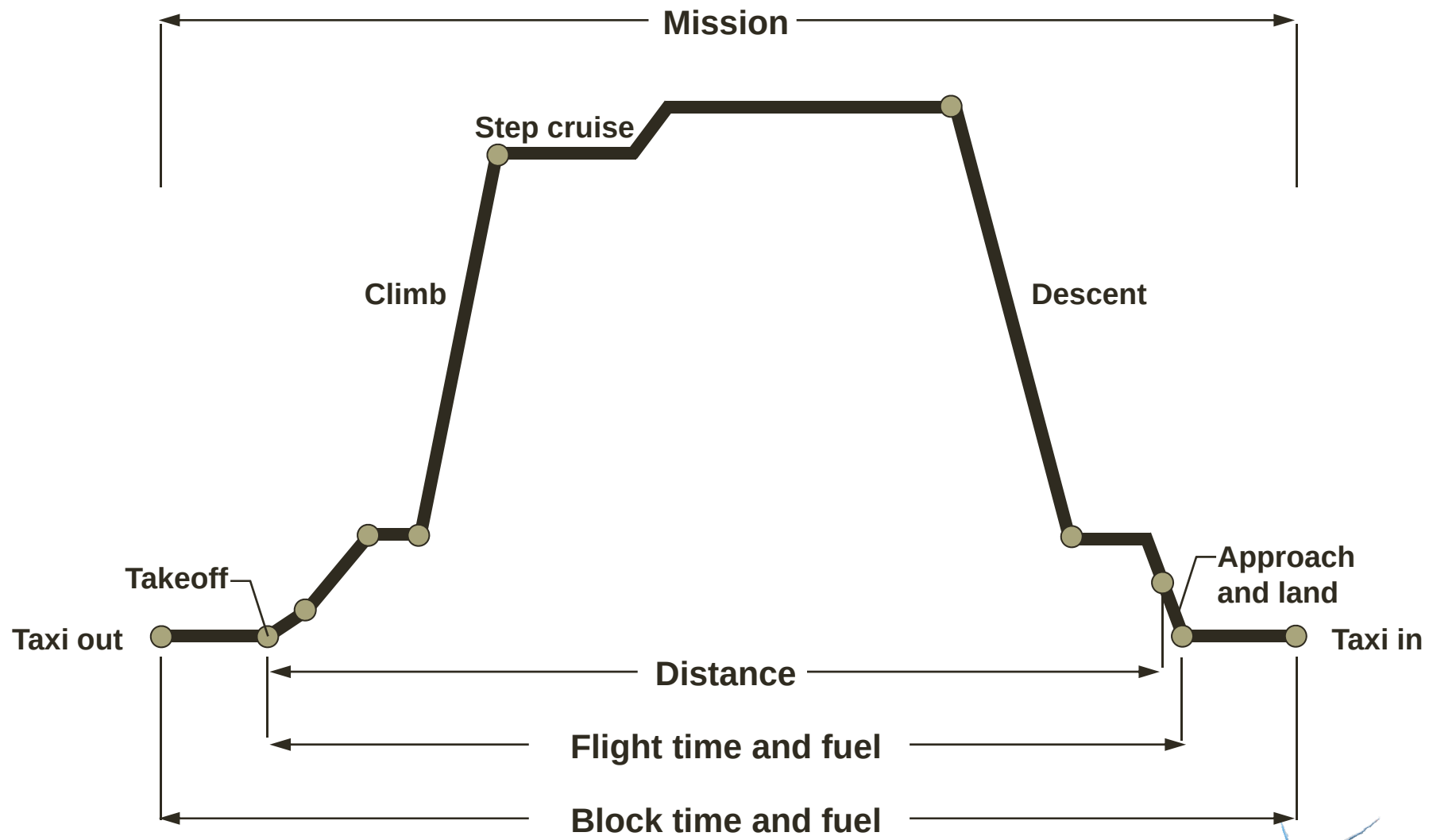
# Airplane Performance

Airplane Characteristics  
Airport Analysis  
**Study Flight Rules**

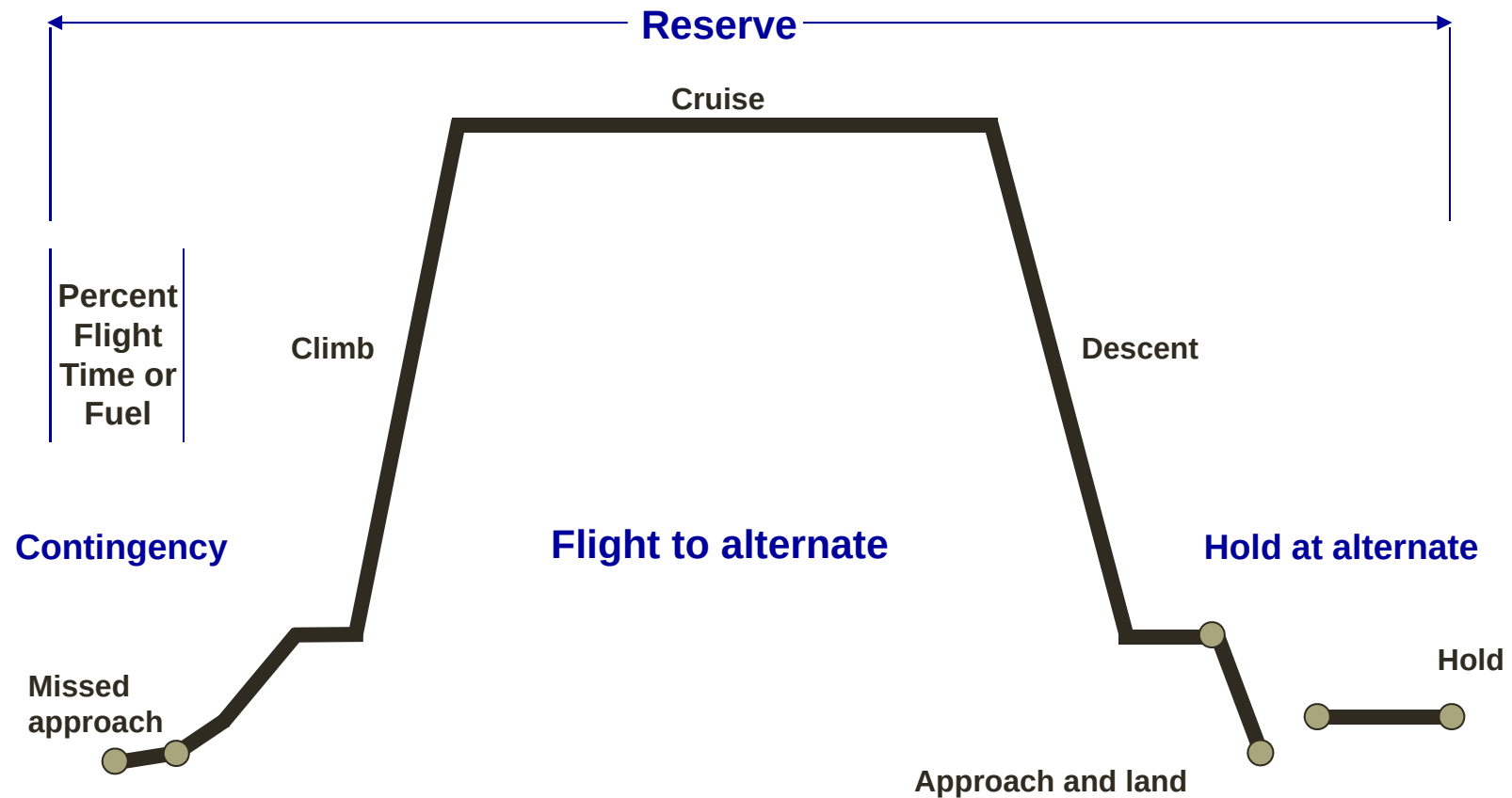
# Study flight rules

- Reserve Fuel Policy: En Route Deviation, Missed Approach; Flight to Alternate Airport; Hold-Over Alternate
- Route Distance
- En Route Winds
- En Route Temperature
- Flight Altitude
- Cruise Procedure
- Alternate Airports

# Study flight rules: typical flight profile

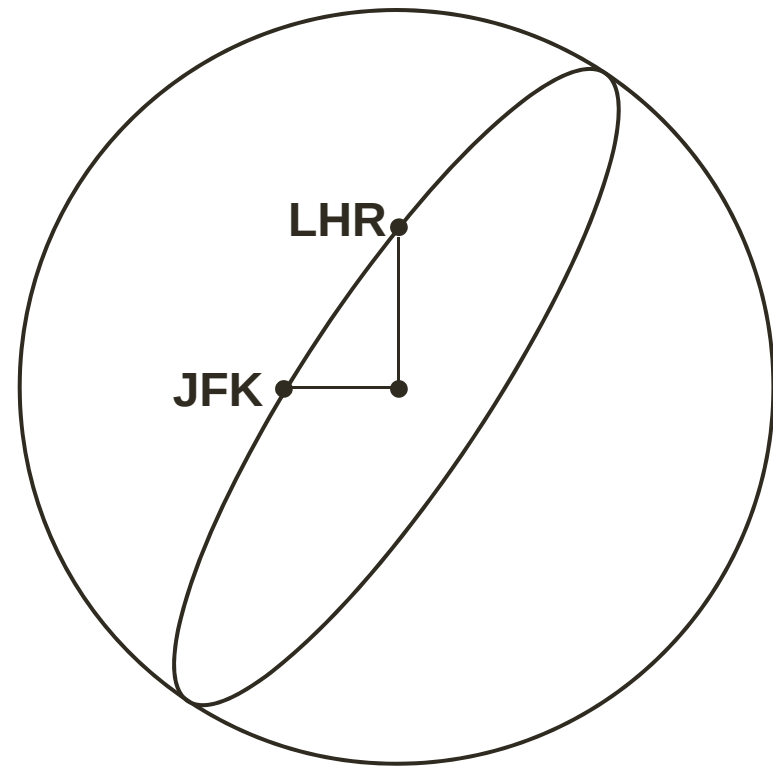


# Study flight rules: typical reserve profile



# Study flight rules: great circle distance

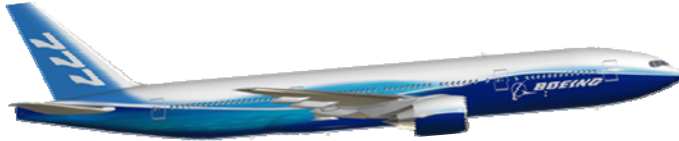
- A plane that passes through the center of the Earth.
- Any arc on the perimeter of this plane will give the great circle distance between the two points that bound the arc.



Copyright © 2013 Boeing. All rights reserved.



# Study flight rules: equivalent still air distance



The **equivalent still air distance** is the distance an airplane would fly in still air on a flight of the same duration as that required to fly the route with a given wind.

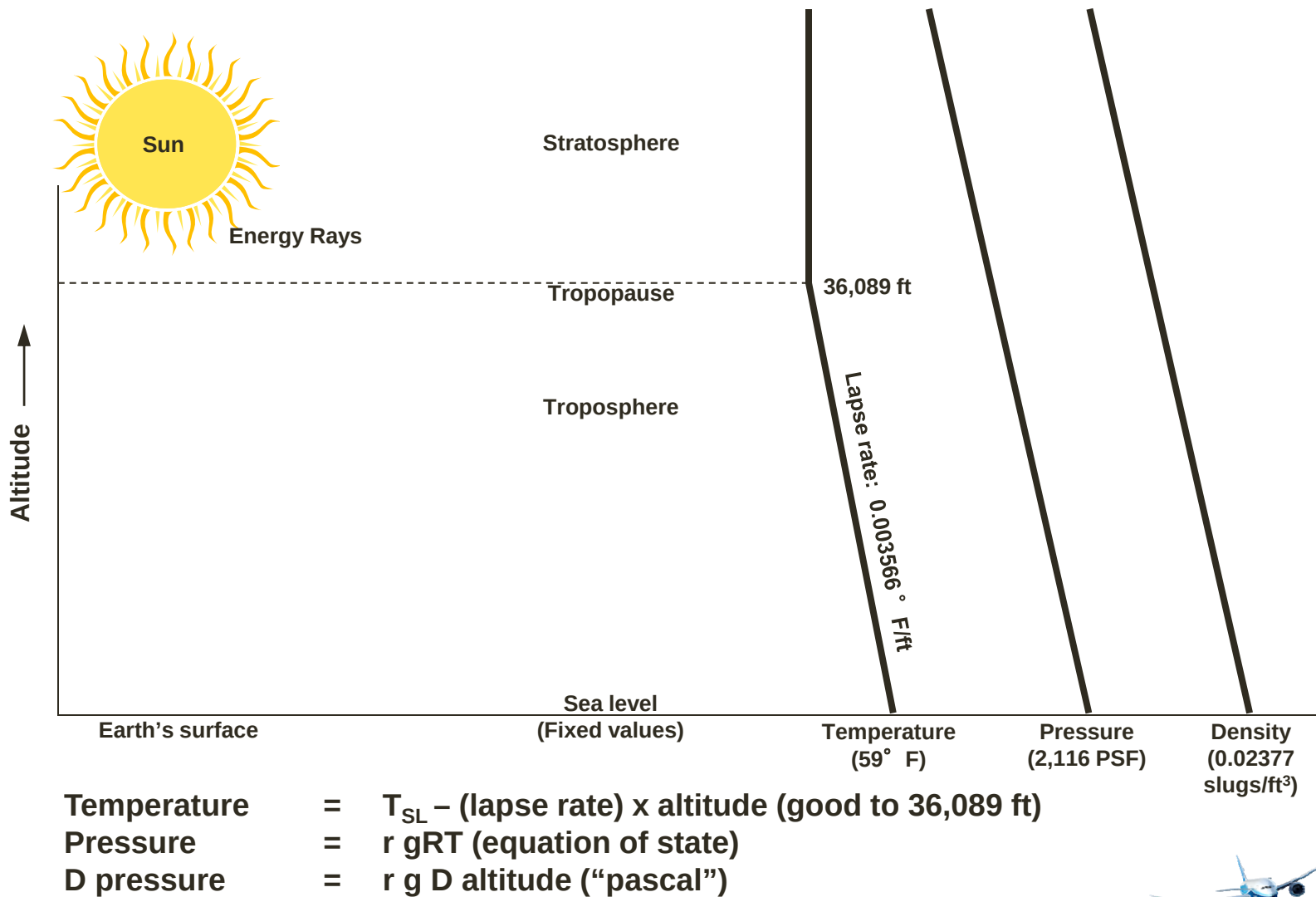
$$\text{ESAD} = \text{distance} \times \frac{(\text{Airplane speed})}{(\text{Airplane speed}) \pm (\text{wind})}$$

**Where:**

|                |   |                                   |
|----------------|---|-----------------------------------|
| ESAD           | = | equivalent still air distance     |
| Distance       | = | airway distance in nautical miles |
| Airplane speed | = | true velocity in knots            |
| Wind           | = | given wind in knots               |



# Study flight rules: International Standard Atmosphere (ISA)

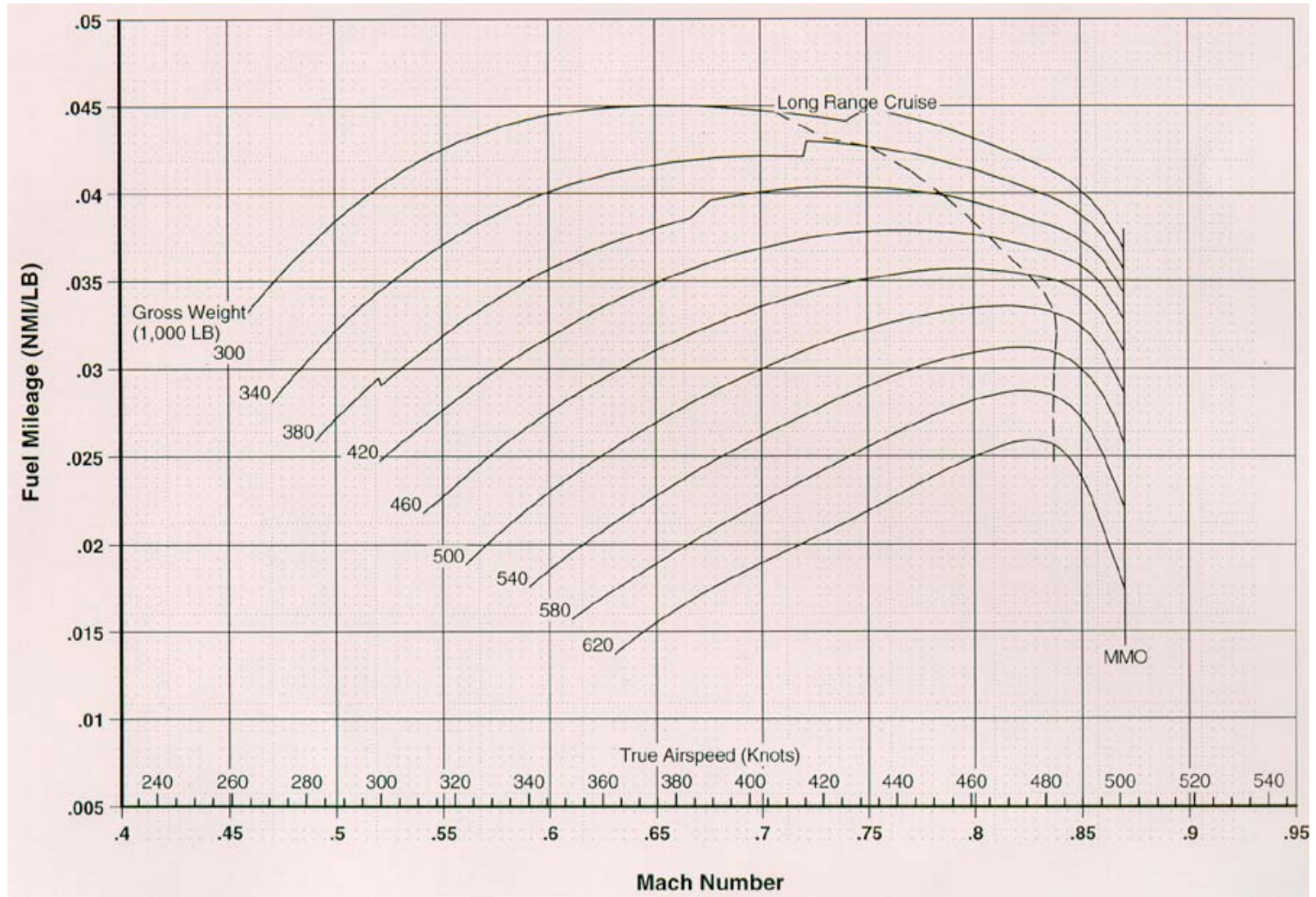


# Airplane Speed

## *Definition of speeds:*

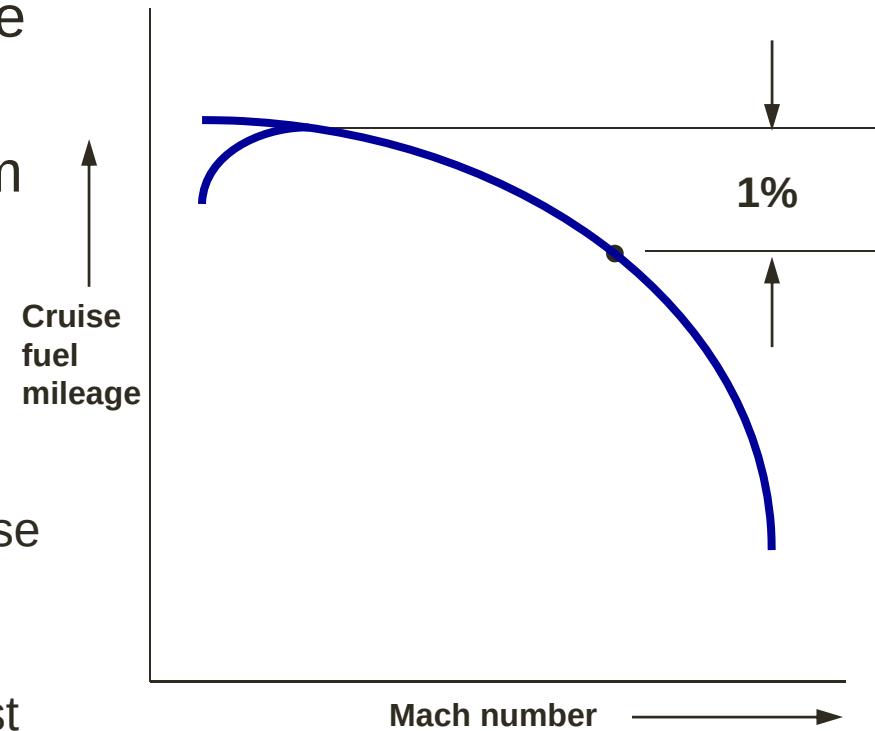
- Airplane speed is described in knots or Mach number
  - 1 knot = 1 nautical mile per hour
  - 1 nautical mile = 6,080 feet = 1,852 meters
  - Mach number is the airplane speed divided by the speed of sound
- Speed schedules
  - MRC ( Maximum Range Cruise) speed schedule that maximizes airplane range
  - LRC (Long Range Cruise) speed schedule for 99% of MRC nams value
  - CI ( Cost Index ) speed schedule relating the cost of flight time to the cost of fuel

# Study flight rules: cruise procedure for fuel mileage, example 777-200 at 35,000 feet, GE 90 engines, standard day



# Study flight rules: cruise procedure for Long-Range Cruise (LRC) speed

- The speed where fuel mileage is 1% less than maximum
- Representative of an optimum cost index speed for most airlines
- Where did LRC originate?
  - Speed stability is difficult to achieve at maximum range cruise speed
  - Before the advent of FMCs, a variable cruise speed (like a cost index) was hard to fly



# Study flight rules: ECON cruise

ECON Cruise Mach = f(Cost Index, CI)

$$CI = \frac{\text{TimeCost}}{\text{FuelCost}} = \frac{\$/\text{hr}}{\text{cents/lb}}$$

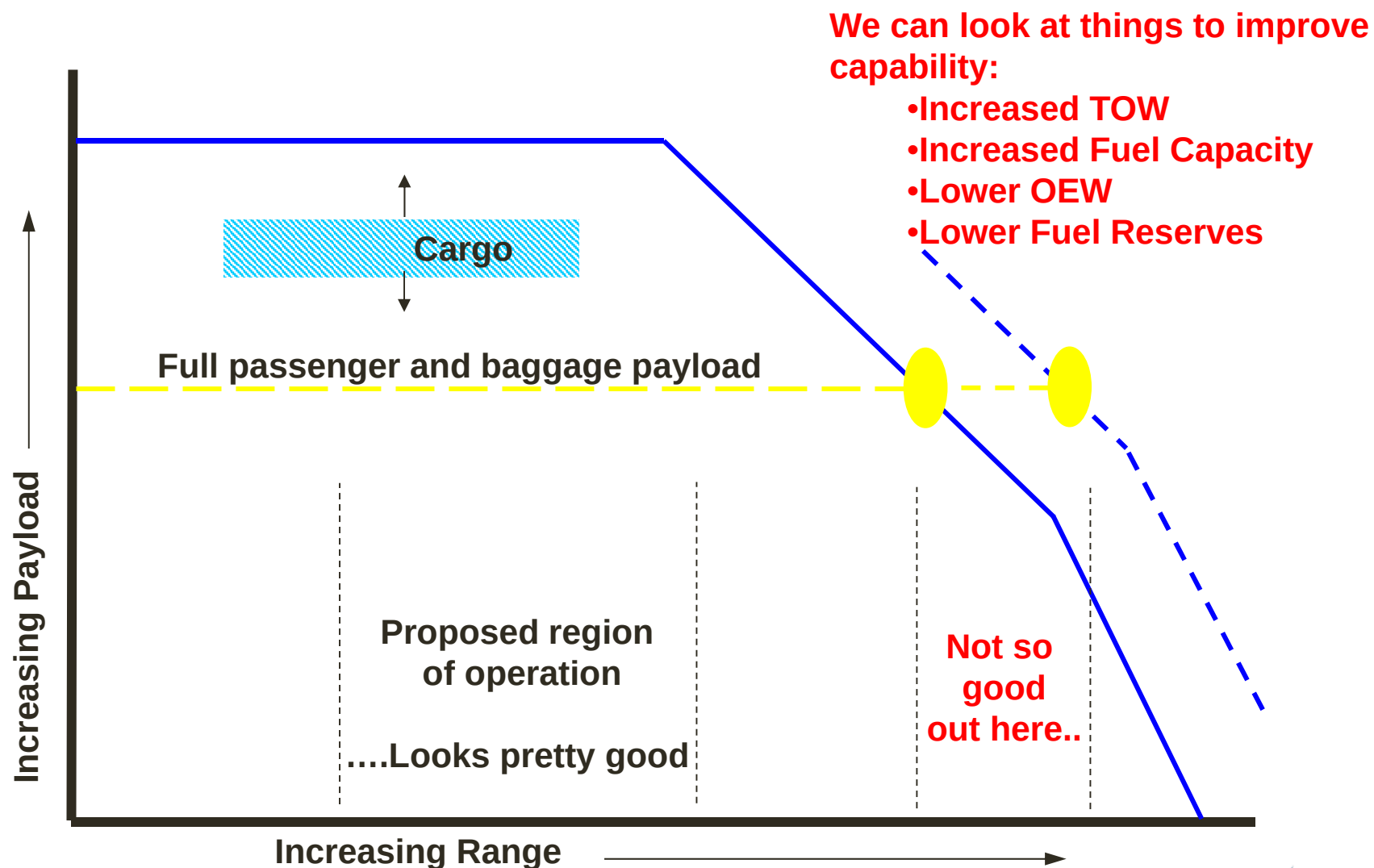
High CI → high speed, high trip fuel, low trip time

Low CI → low speed, low fuel burn, high trip time

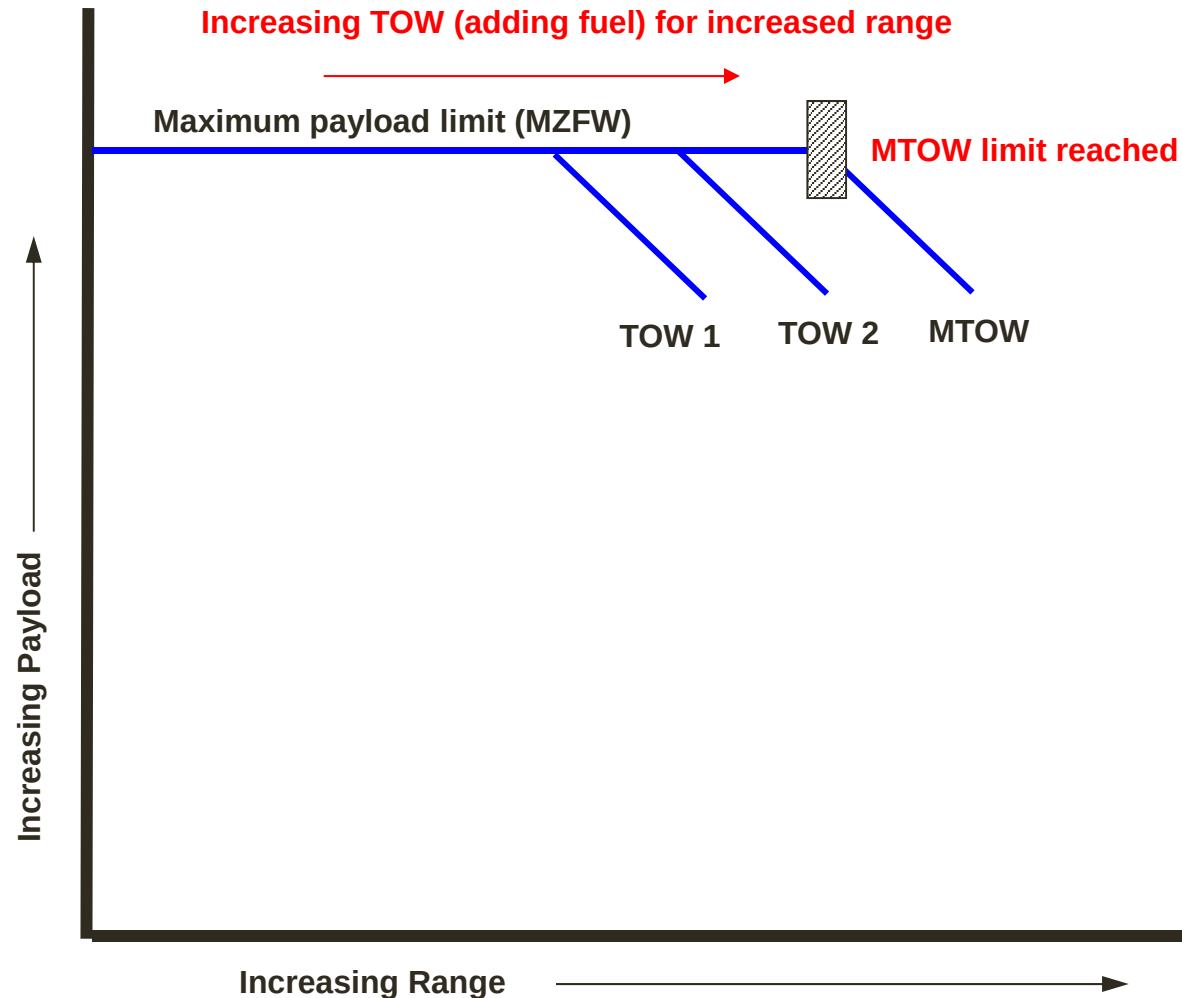
# Study flight rules: factors affecting fuel burn

- Range                      Increasing range increases fuel burn
- Weight                     Increasing weight increases fuel burn
  - Operating Empty Weight
  - Payload
  - Reserve fuel policy
- Cruise Speed             Cruise speed above optimum increases fuel burn
- Altitude                    Higher cruise altitude decreases fuel burn
- Enroute Wind             Headwind increases fuel burn; tailwind decreases fuel burn
- Enroute Temperature    Higher en route temperature increases fuel burn

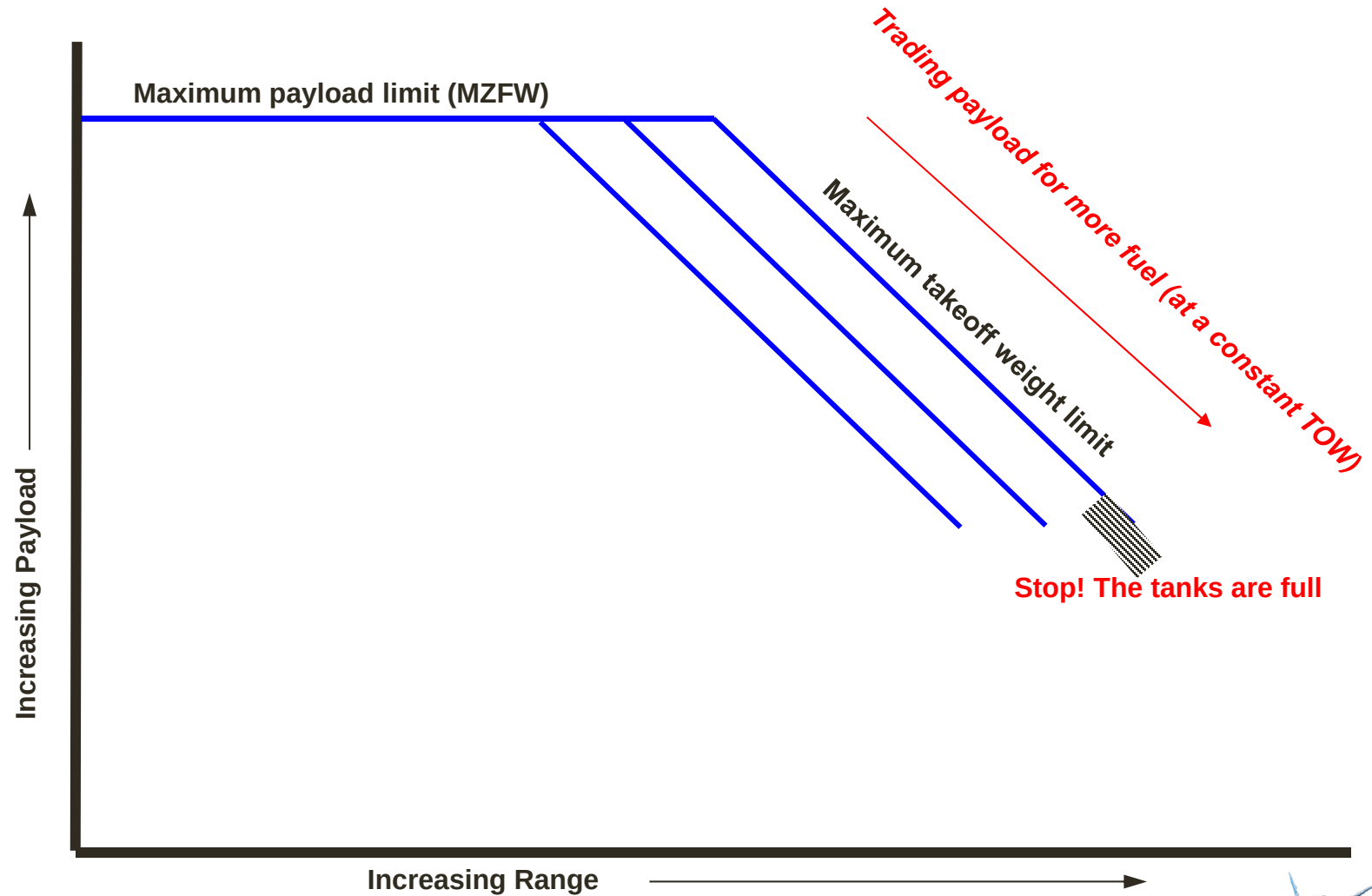
# Study flight rules: usefulness of payload range curve



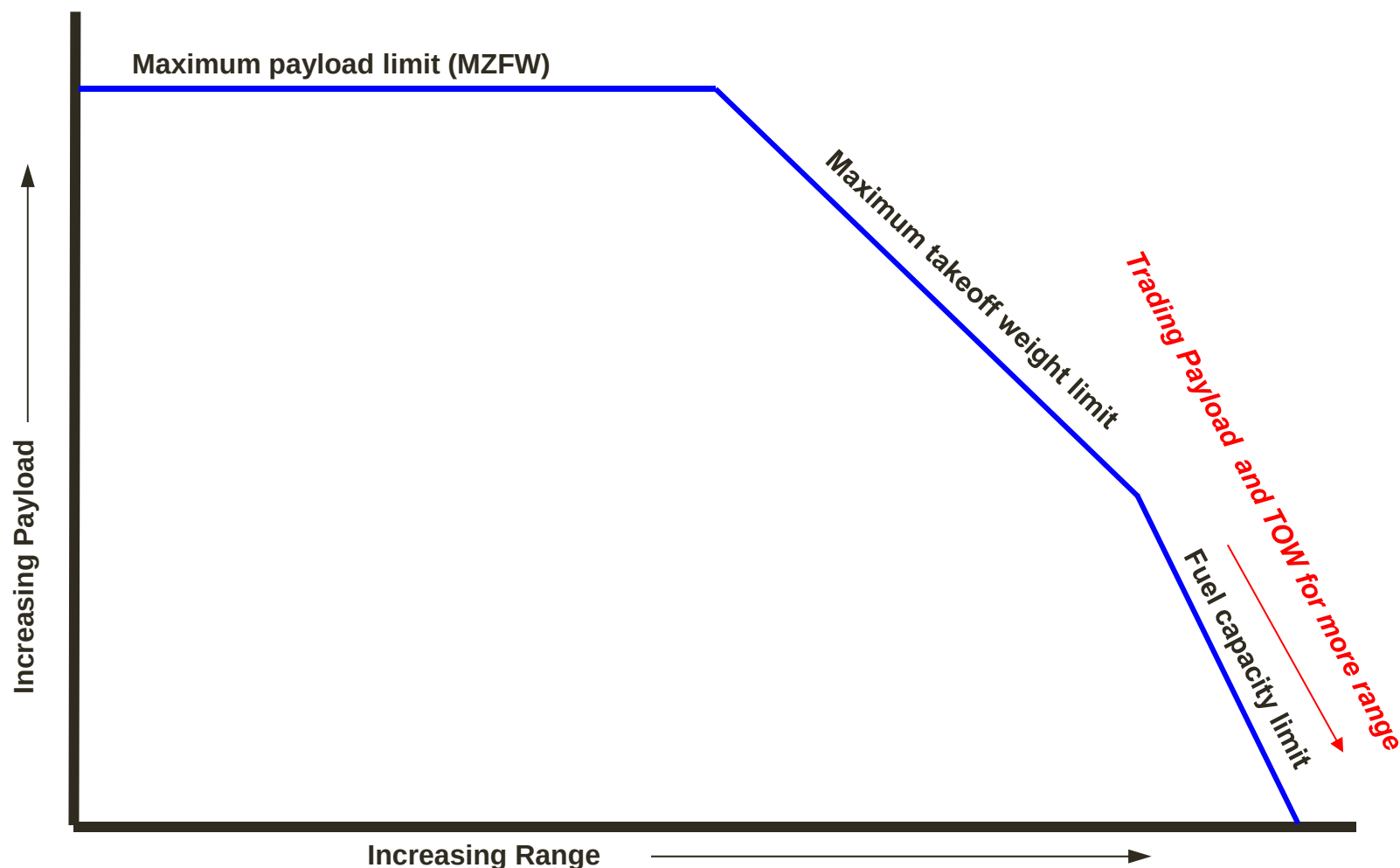
# Study flight rules: payload vs. range curve evolution



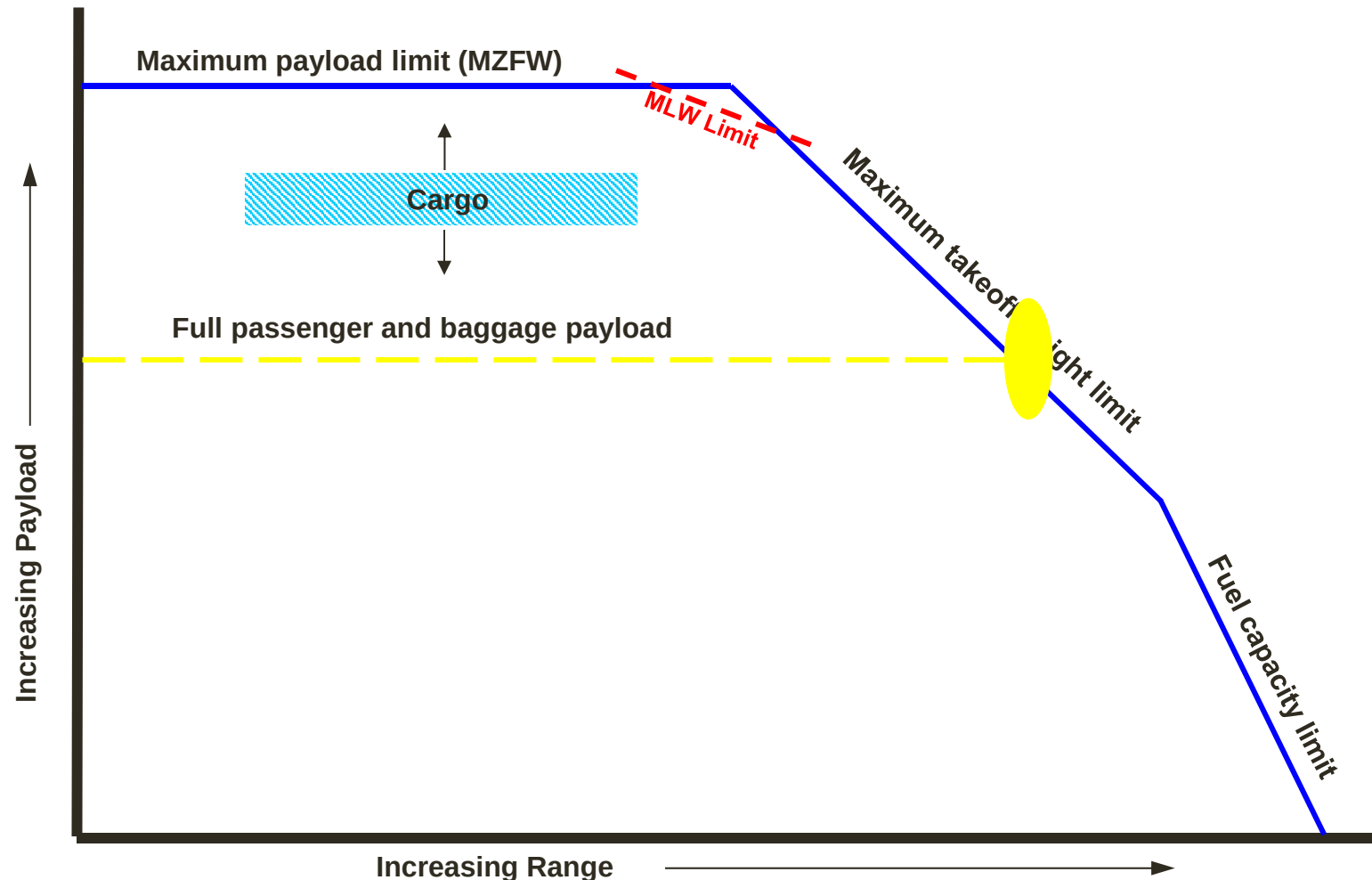
# Study flight rules: payload vs. range curve evolution



# Study flight rules: payload vs. range curve evolution



# Study flight rules: payload vs. range curve – A complete performance envelope for the aircraft



# Study flight rules: “circle chart” range capability from Denver

## *Full Passenger Payload*

### **737-700W/CFM56-7B22E Basic**

Payload: 26,460 lb  
MTOW: 133,000 lb OEW: 85,090 lb  
Fuel Capacity: 6,875 gal  
Passengers: 126 (1,374 nm)

### **737-700W/CFM56-7B22E HGW**

Payload: 26,460 lb  
MTOW: 154,500 lb OEW: 85,090 lb  
Fuel Capacity: 6,875 gal  
Passengers: 126 (3,173 nm)

### Boeing Typical Rules

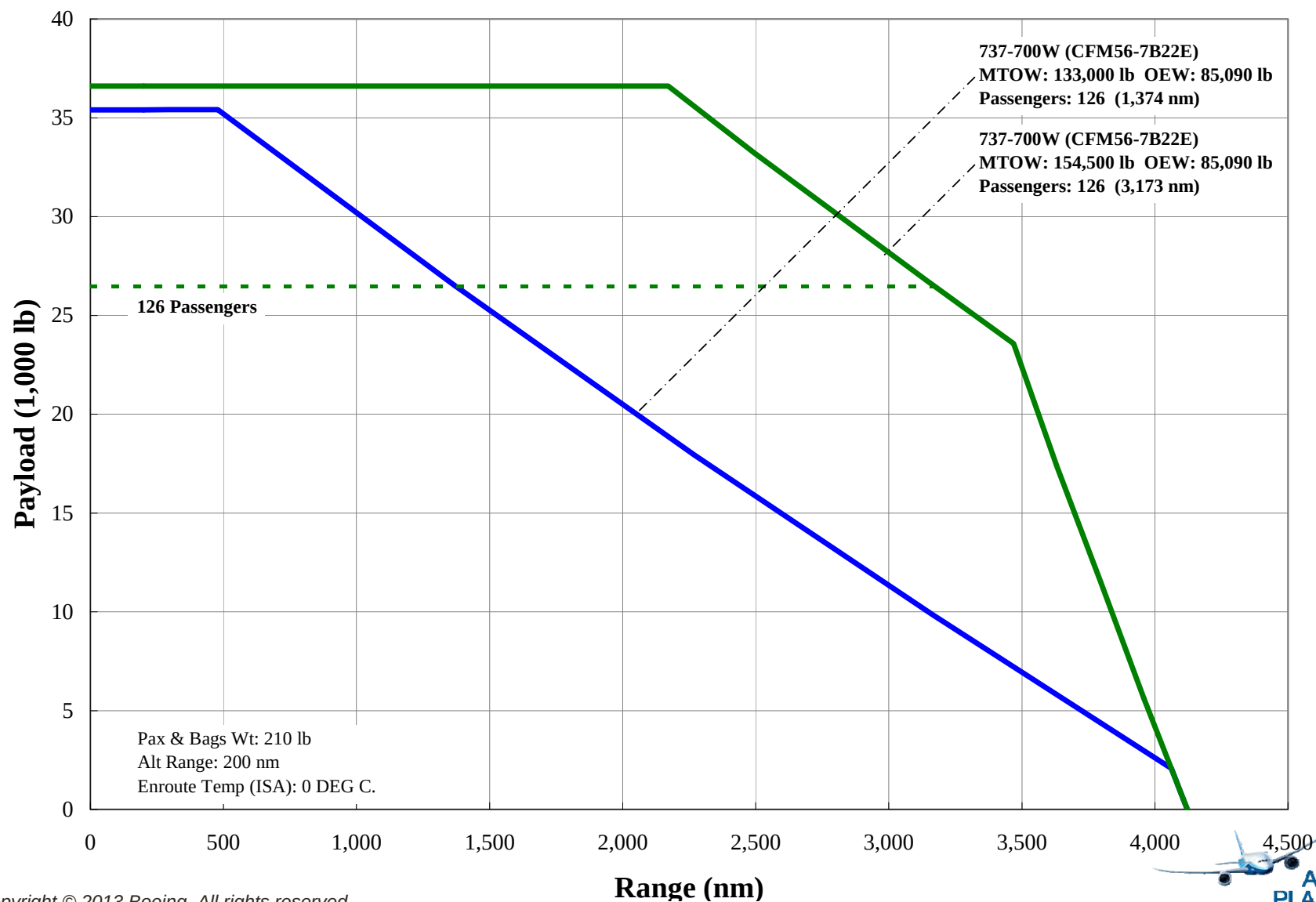
85% Annual Winds  
2% Airway Allowance  
2% Fuel Burn Factor  
210 lb per Pax and Bags  
200 nm alternate



# Payload range chart

## 737-700W Gross Weight Comparison Payload - Range Capability

A



# Payload range data – high gross weight

## Mission Performance Summary

|                                                                              |        |           |                    |                   |                        |                   |           |
|------------------------------------------------------------------------------|--------|-----------|--------------------|-------------------|------------------------|-------------------|-----------|
| <b>737-700W Gross Weight Comparison</b>                                      |        |           | Max Taxi Wt:       | 155,000 lb        | Structural Payload:    | 36,610 lb         |           |
| <b>Typical Intl Rules; 2% Fuel Markup</b>                                    |        |           | Max TO Wt:         | 154,500 lb        | Study Payload:         | 36,610 lb         |           |
| <b>Payload Range Data</b>                                                    |        |           | Max Land Wt:       | 129,200 lb        | Passenger Seats:       | 126 @ 210 lb ea   |           |
| Enroute Wind Reliability:                                                    | 85%    | Airframe: | <b>737-700W</b>    | Max Zero Fuel Wt: | 121,700 lb             | Pax Bags Payload: | 26,460 lb |
| Enroute Wind & Temps Season:                                                 | Annual | Engine:   | <b>CFM56-7B22E</b> | Op Empty Wt:      | 85,090 lb              | Rev Cargo:        | 10,150 lb |
|                                                                              |        | Interior: | <b>126</b>         | Fuel Capacity:    | 6,875 gal              |                   |           |
|                                                                              |        |           |                    | Fuel Wt:          | 46,063 lb @ 6.7 lb/gal |                   |           |
| APS job.APNav / 737-700W / CFM56-7B22E / D6388275REVA / 8/30/2011 1:01:01 PM |        |           |                    |                   |                        |                   |           |

| Track                | Rwy | Dist<br>(nm) | ESAD<br>(nm) | Cruise<br>Altitude<br>(100ft) | Cruise<br>Wind<br>(kt) | Cruise<br>Delta<br>Temp<br>(°C) | Mach | Block<br>Time<br>(h) | Takeoff<br>Weight<br>(lb) | Landing<br>Weight<br>(lb) | Block<br>Fuel<br>(lb) | Reserve<br>Fuel<br>(lb) | PAX | Cargo<br>(lb) | Payload<br>(lb) | Altn | Altn<br>Dist<br>(nm) |
|----------------------|-----|--------------|--------------|-------------------------------|------------------------|---------------------------------|------|----------------------|---------------------------|---------------------------|-----------------------|-------------------------|-----|---------------|-----------------|------|----------------------|
| <b>Payload Range</b> |     |              |              |                               |                        |                                 |      |                      |                           |                           |                       |                         |     |               |                 |      |                      |
| ORIG to DEST         | -   | 200          | 200          | 200                           | 0                      | 0                               | LRC  | 0.928                | 131,295                   | 127,596                   | 4,056                 | 5,896                   | 126 | 10,150        | 36,610          | ALTN | 200                  |
| ORIG to DEST         | -   | 857          | 857          | 380                           | 0                      | 0                               | LRC  | 2.375                | 138,571                   | 127,940                   | 10,988                | 6,241                   | 126 | 10,149        | 36,610          | ALTN | 200                  |
| ORIG to DEST         | -   | 1,514        | 1,514        | 380                           | 0                      | 0                               | LRC  | 3.839                | 146,329                   | 128,309                   | 18,377                | 6,611                   | 126 | 10,148        | 36,610          | ALTN | 200                  |
| ORIG to DEST         | -   | 2,171        | 2,171        | 360/380                       | 0/0                    | 0/0                             | LRC  | 5.310                | 154,500                   | 128,701                   | 26,156                | 7,001                   | 126 | 10,150        | 36,610          | ALTN | 200                  |
| ORIG to DEST         | -   | 2,496        | 2,496        | 360/380/400                   | 0/0/0                  | 0/0/0                           | LRC  | 6.026                | 154,500                   | 125,376                   | 29,481                | 7,074                   | 126 | 6,752         | 33,212          | ALTN | 200                  |
| ORIG to DEST         | -   | 2,820        | 2,820        | 360/380/400                   | 0/0/0                  | 0/0/0                           | LRC  | 6.747                | 154,500                   | 122,152                   | 32,705                | 7,090                   | 126 | 3,512         | 29,972          | ALTN | 200                  |
| ORIG to DEST         | -   | 3,144        | 3,144        | 360/380/400                   | 0/0/0                  | 0/0/0                           | LRC  | 7.471                | 154,500                   | 118,994                   | 35,863                | 7,161                   | 126 | 283           | 26,743          | ALTN | 200                  |
| ORIG to DEST         | -   | 3,173        | 3,173        | 360/380/400                   | 0/0/0                  | 0/0/0                           | LRC  | 7.535                | 154,500                   | 118,718                   | 36,139                | 7,168                   | 126 | 0             | 26,460          | ALTN | 200                  |
| ORIG to DEST         | -   | 3,469        | 3,469        | 360/380/400                   | 0/0/0                  | 0/0/0                           | LRC  | 8.198                | 154,500                   | 115,897                   | 38,960                | 7,230                   | 112 | 0             | 23,577          | ALTN | 200                  |
| ORIG to DEST         | -   | 3,632        | 3,632        | 360/380/400                   | 0/0/0                  | 0/0/0                           | LRC  | 8.577                | 148,263                   | 109,497                   | 39,124                | 7,067                   | 82  | 0             | 17,340 FUEL     | ALTN | 200                  |
| ORIG to DEST         | -   | 3,795        | 3,795        | 380/400                       | 0/0                    | 0/0                             | LRC  | 8.975                | 142,455                   | 103,513                   | 39,299                | 6,893                   | 54  | 0             | 11,530 FUEL     | ALTN | 200                  |
| ORIG to DEST         | -   | 3,958        | 3,958        | 380/400                       | 0/0                    | 0/0                             | LRC  | 9.389                | 136,555                   | 97,464                    | 39,448                | 6,742                   | 26  | 0             | 5,632 FUEL      | ALTN | 200                  |
| ORIG to DEST         | -   | 4,121        | 4,121        | 400                           | 0                      | 0                               | LRC  | 9.828                | 130,923                   | 91,670                    | 39,610                | 6,580                   | 0   | 0             | 0 FUEL          | ALTN | 200                  |

# Payload range data – basic gross weight

## Mission Performance Summary

|                                                                              |           |                    |                   |                        |                     |                 |
|------------------------------------------------------------------------------|-----------|--------------------|-------------------|------------------------|---------------------|-----------------|
| <b>737-700W Gross Weight Comparison</b>                                      |           |                    | Max Taxi Wt:      | 133,500 lb             | Structural Payload: | 35,410 lb       |
| <b>Typical Intl Rules; 2% Fuel Markup</b>                                    |           |                    | Max TO Wt:        | 133,000 lb             | Study Payload:      | 35,410 lb       |
| <b>Payload Range Data</b>                                                    |           |                    | Max Land Wt:      | 128,000 lb             | Passenger Seats:    | 126 @ 210 lb ea |
| Enroute Wind Reliability: 85%                                                | Airframe: | <b>737-700W</b>    | Max Zero Fuel Wt: | 120,500 lb             | Pax Bags Payload:   | 26,460 lb       |
| Enroute Wind & Temps Season: Annual                                          | Engine:   | <b>CFM56-7B22E</b> | Op Empty Wt:      | 85,090 lb              | Rev Cargo:          | 8,950 lb        |
|                                                                              | Interior: | <b>126</b>         | Fuel Capacity:    | 6,875 gal              |                     |                 |
|                                                                              |           |                    | Fuel Wt:          | 46,063 lb @ 6.7 lb/gal |                     |                 |
| APS job.APNav / 737-700W / CFM56-7B22E / D6388275REVA / 8/30/2011 1:01:00 PM |           |                    |                   |                        |                     |                 |

| Track                | Rwy | Dist<br>(nm) | ESAD<br>(nm) | Cruise<br>Altitude<br>(100ft) | Cruise<br>Wind<br>(kt) | Cruise<br>Delta<br>Temp<br>(°C) | Mach | Block<br>Time<br>(h) | Takeoff<br>Weight<br>(lb) | Landing<br>Weight<br>(lb) | Block<br>Fuel<br>(lb) | Reserve<br>Fuel<br>(lb) | PAX | Cargo<br>(lb) | Payload<br>(lb) | Altn | Altn<br>Dist<br>(nm) |
|----------------------|-----|--------------|--------------|-------------------------------|------------------------|---------------------------------|------|----------------------|---------------------------|---------------------------|-----------------------|-------------------------|-----|---------------|-----------------|------|----------------------|
| <b>Payload Range</b> |     |              |              |                               |                        |                                 |      |                      |                           |                           |                       |                         |     |               |                 |      |                      |
| ORIG to DEST         | -   | 200          | 200          | 200                           | 0                      | 0                               | LRC  | 0.929                | 130,041                   | 126,360                   | 4,038                 | 5,863                   | 126 | 8,947         | 35,410          | ALTN | 200                  |
| ORIG to DEST         | -   | 293          | 293          | 260                           | 0                      | 0                               | LRC  | 1.151                | 131,110                   | 126,413                   | 5,054                 | 5,913                   | 126 | 8,950         | 35,410          | ALTN | 200                  |
| ORIG to DEST         | -   | 386          | 386          | 320                           | 0                      | 0                               | LRC  | 1.340                | 132,118                   | 126,461                   | 6,014                 | 5,961                   | 126 | 8,950         | 35,410          | ALTN | 200                  |
| ORIG to DEST         | -   | 479          | 479          | 380                           | 0                      | 0                               | LRC  | 1.527                | 133,000                   | 126,503                   | 6,854                 | 6,003                   | 126 | 8,950         | 35,410          | ALTN | 200                  |
| ORIG to DEST         | -   | 1,374        | 1,374        | 400                           | 0                      | 0                               | LRC  | 3.513                | 133,000                   | 117,694                   | 15,663                | 6,144                   | 126 | 0             | 26,460          | ALTN | 200                  |
| ORIG to DEST         | -   | 1,374        | 1,374        | 400                           | 0                      | 0                               | LRC  | 3.514                | 133,000                   | 117,688                   | 15,669                | 6,144                   | 125 | 0             | 26,454          | ALTN | 200                  |
| ORIG to DEST         | -   | 2,270        | 2,270        | 400                           | 0                      | 0                               | LRC  | 5.534                | 133,000                   | 109,350                   | 24,007                | 6,327                   | 85  | 0             | 17,932          | ALTN | 200                  |
| ORIG to DEST         | -   | 3,166        | 3,166        | 400                           | 0                      | 0                               | LRC  | 7.581                | 133,000                   | 101,410                   | 31,947                | 6,480                   | 46  | 0             | 9,841           | ALTN | 200                  |
| ORIG to DEST         | -   | 4,062        | 4,062        | 400                           | 0                      | 0                               | LRC  | 9.661                | 133,000                   | 93,799                    | 39,558                | 6,632                   | 9   | 0             | 2,077           | ALTN | 200                  |
| ORIG to DEST         | -   | 4,076        | 4,076        | 400                           | 0                      | 0                               | LRC  | 9.703                | 132,475                   | 93,262                    | 39,569                | 6,619                   | 7   | 0             | 1,553 FUEL      | ALTN | 200                  |
| ORIG to DEST         | -   | 4,091        | 4,091        | 400                           | 0                      | 0                               | LRC  | 9.744                | 131,960                   | 92,733                    | 39,583                | 6,606                   | 4   | 0             | 1,037 FUEL      | ALTN | 200                  |
| ORIG to DEST         | -   | 4,106        | 4,106        | 400                           | 0                      | 0                               | LRC  | 9.786                | 131,446                   | 92,206                    | 39,598                | 6,593                   | 2   | 0             | 523 FUEL        | ALTN | 200                  |
| ORIG to DEST         | -   | 4,121        | 4,121        | 400                           | 0                      | 0                               | LRC  | 9.828                | 130,923                   | 91,670                    | 39,610                | 6,580                   | 0   | 0             | 0 FUEL          | ALTN | 200                  |

# Key takeaways

- Airplane Characteristics

+

- Airport Analysis

+

- Study Flight Rules



- Range
- Payload
- Takeoff Performance
- Block Time and Block Fuel

# MonteCristoAir case study connection

- Aircraft performance capability can significantly impact the strategy MonteCristoAir chooses to respond to competition

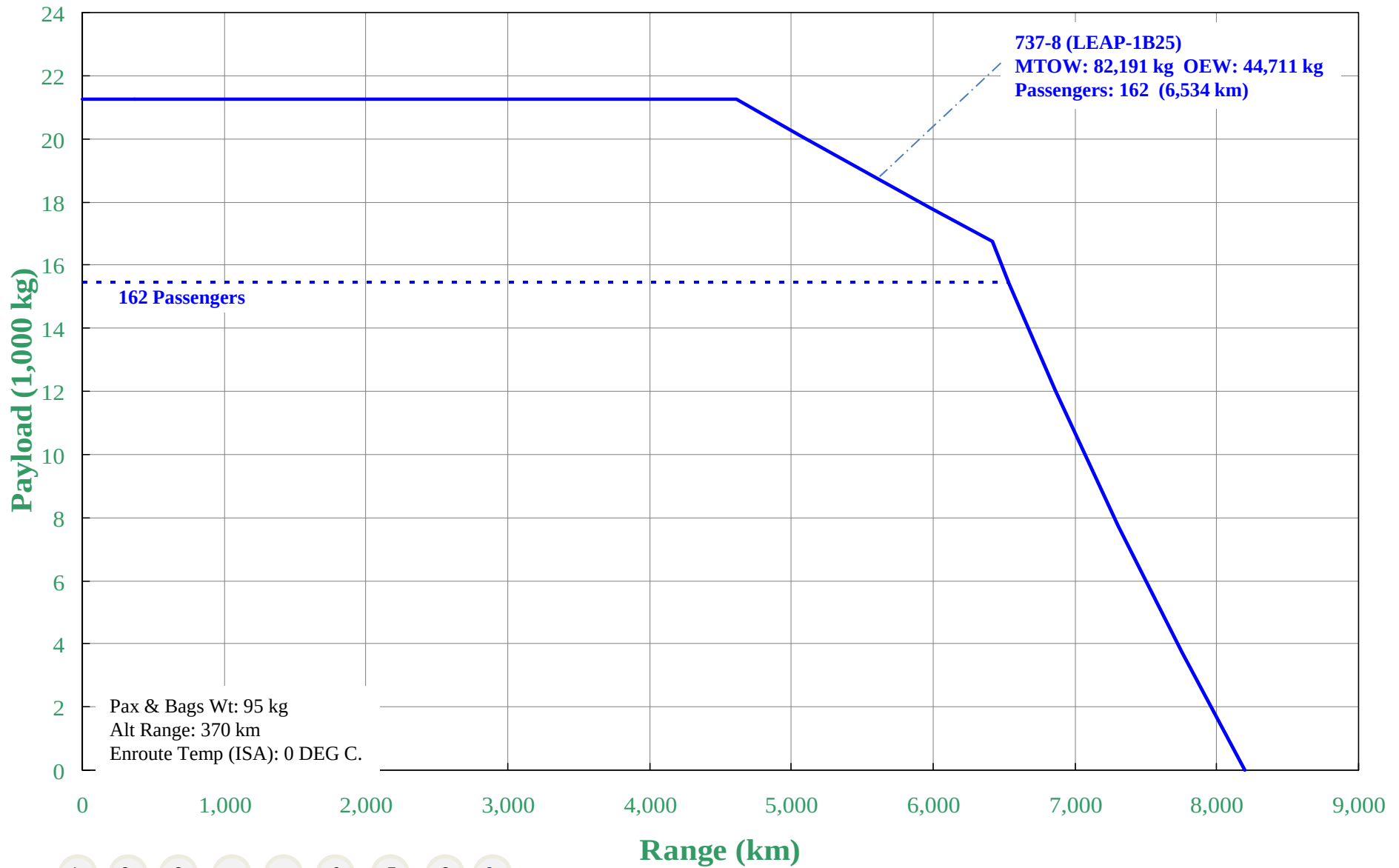
# Payload Range Questions:

1. What would be the effect of an OEW increase?
2. How would a passenger weight increase affect design range?
3. Where would an airport takeoff limit show up on a payload range curve?
4. How would a very long flight to alternate distance affect design range?
5. How would a hot enroute cruise temperature affect design range?
6. How does an auxiliary fuel tank affect the design range? (This has a two part answer)
7. How does reducing reserves contingency fuel affect a payload range?
8. How is a payload range curve impacted by flying at off optimum altitudes?
9. How does speed affect a design range?

# MonteCristoAir

## Payload - Range Capability

A



1

2

3

4

5

6

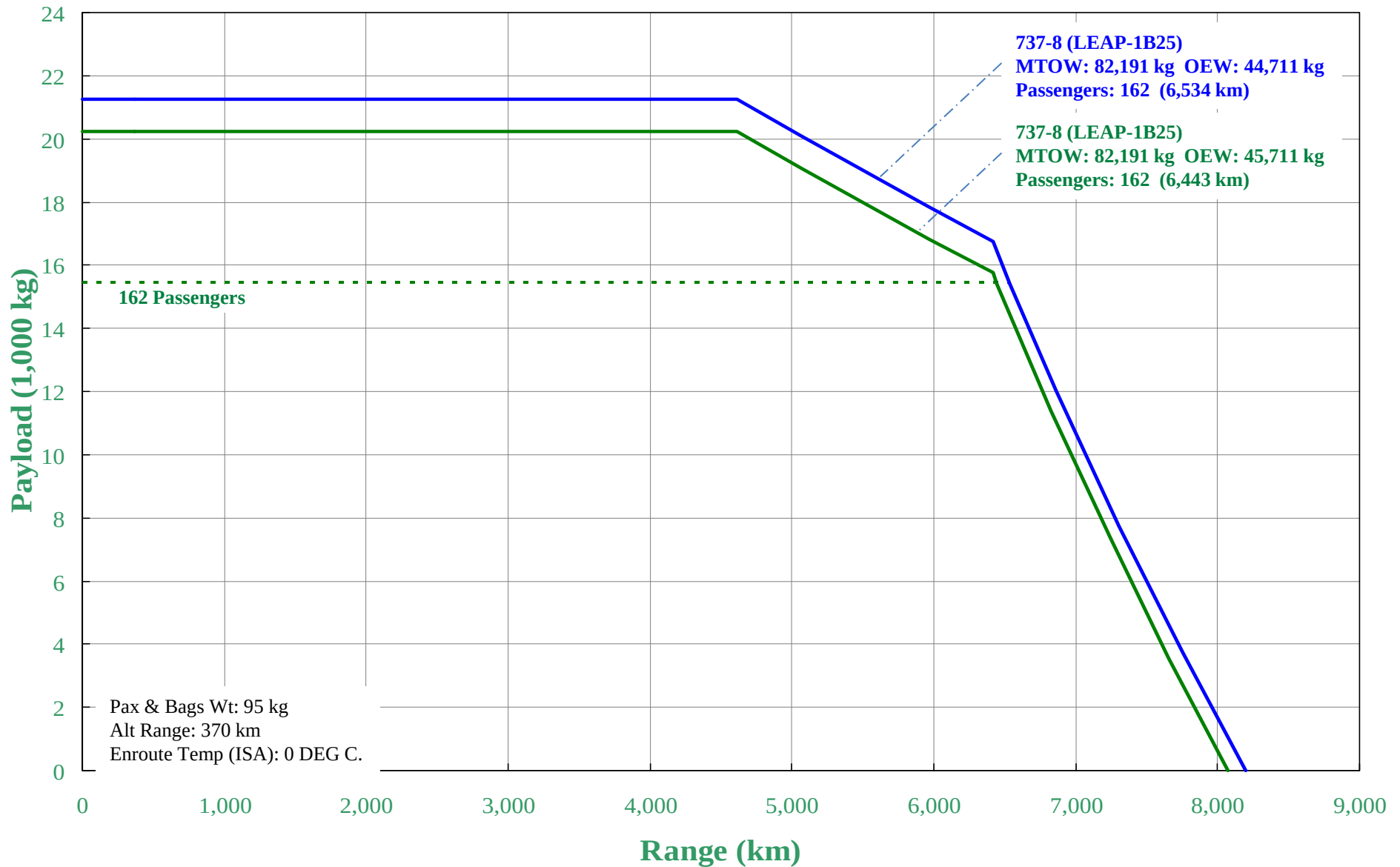
7

8

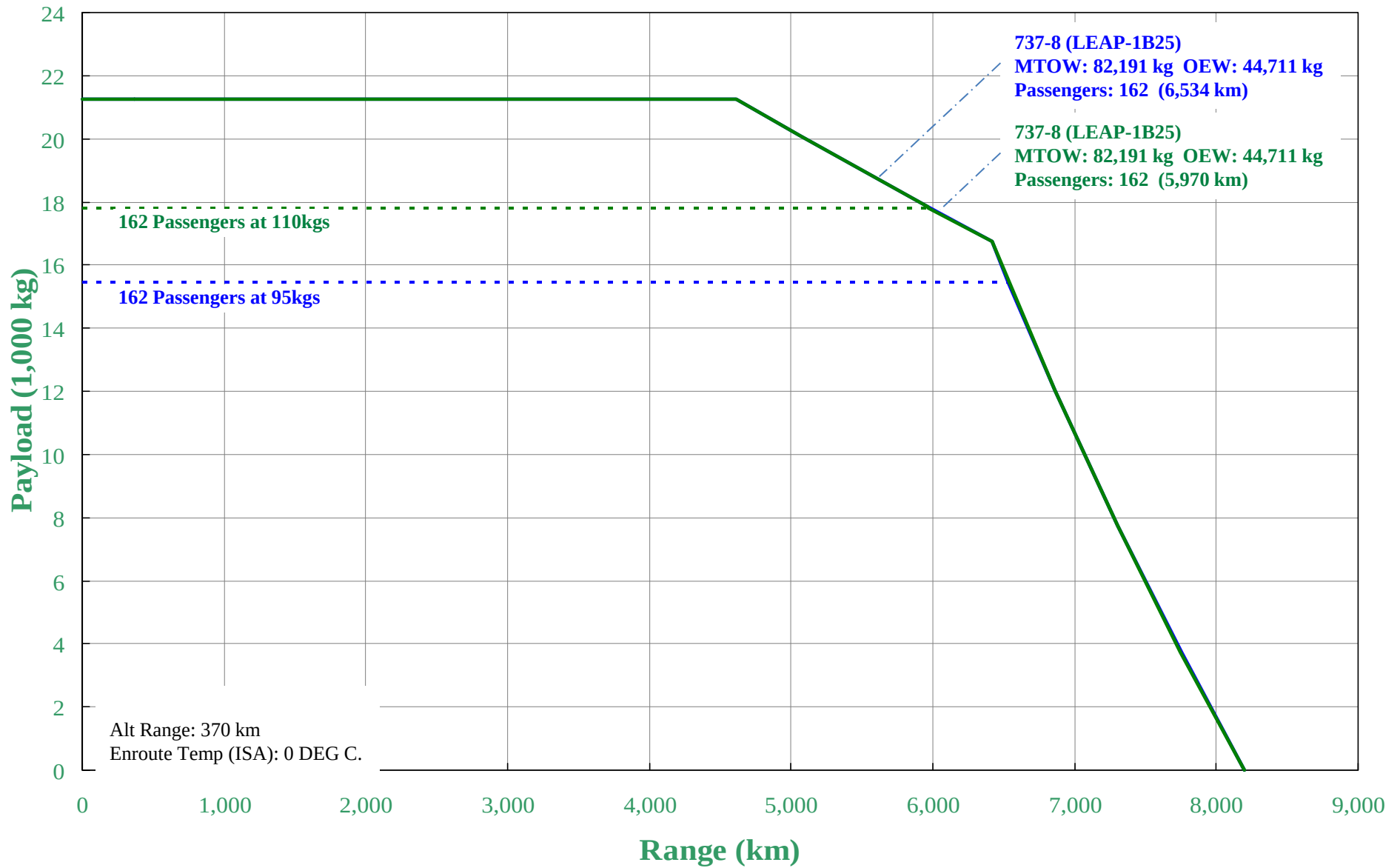
9

MonteCristoAir

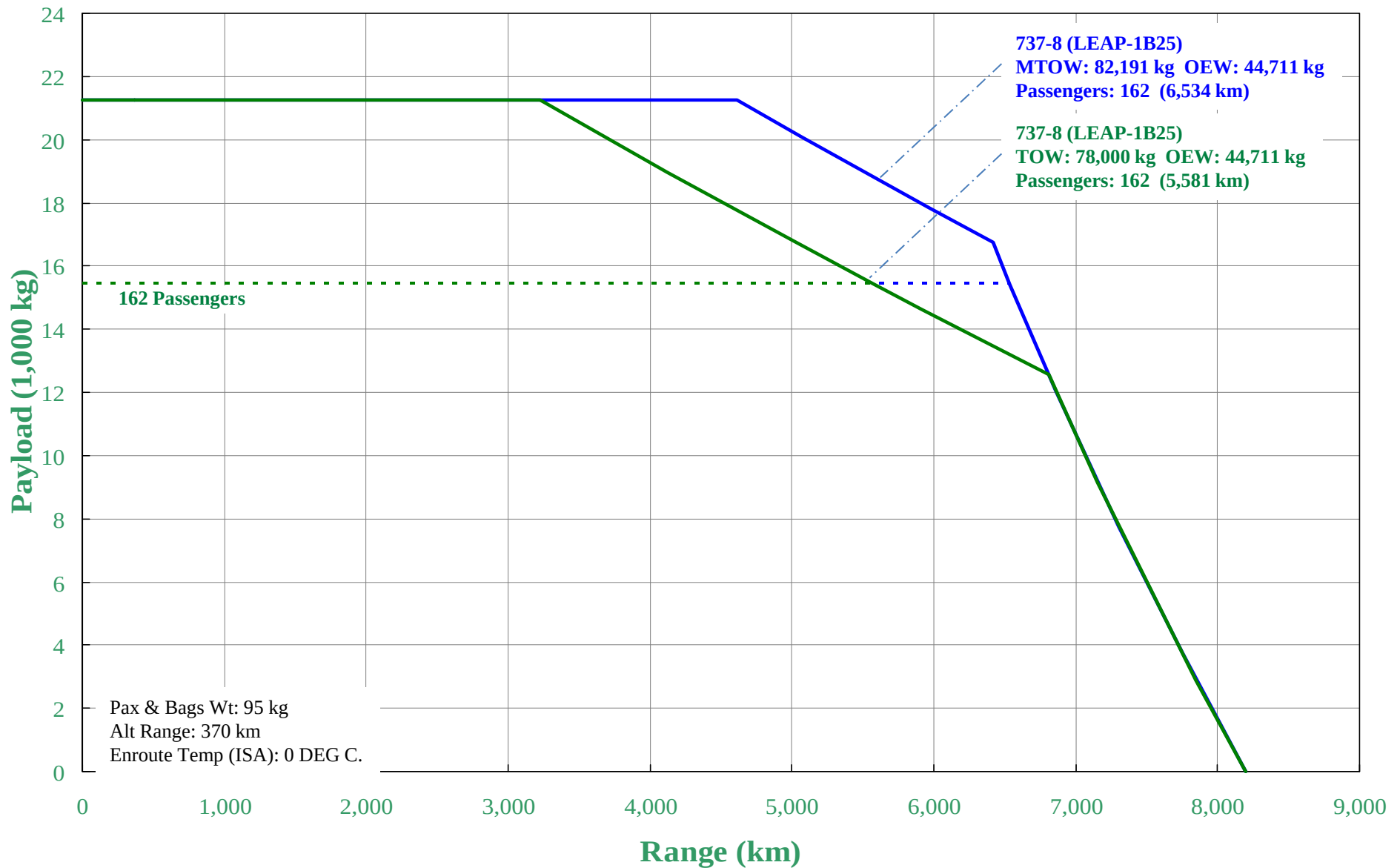
# Payload - Range Capability - Effect of OEW Increase



# Payload - Range Capability - Effect of Pax Wt Increase



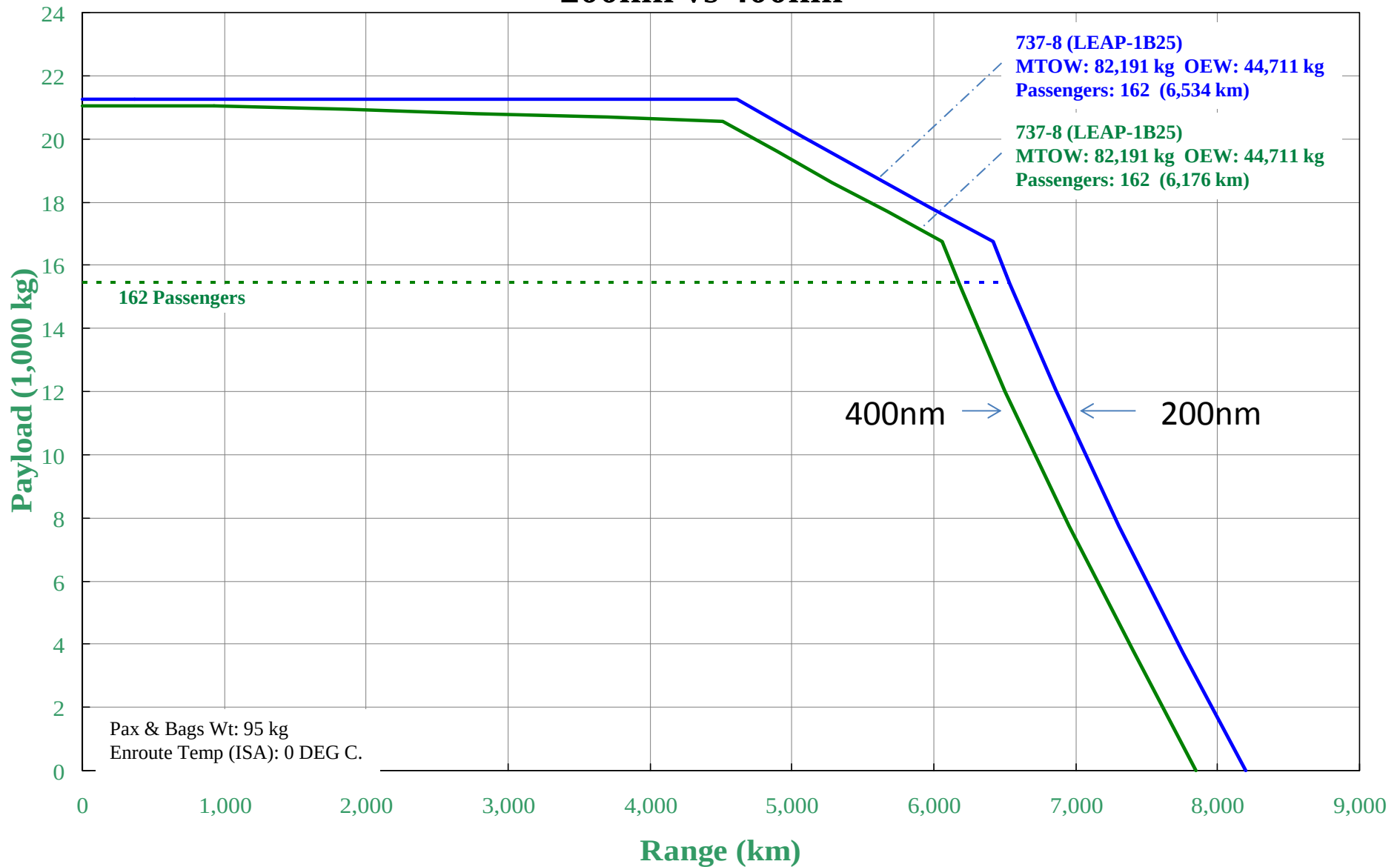
## Payload - Range Capability - Effect of Airport Takeoff Weight Limit



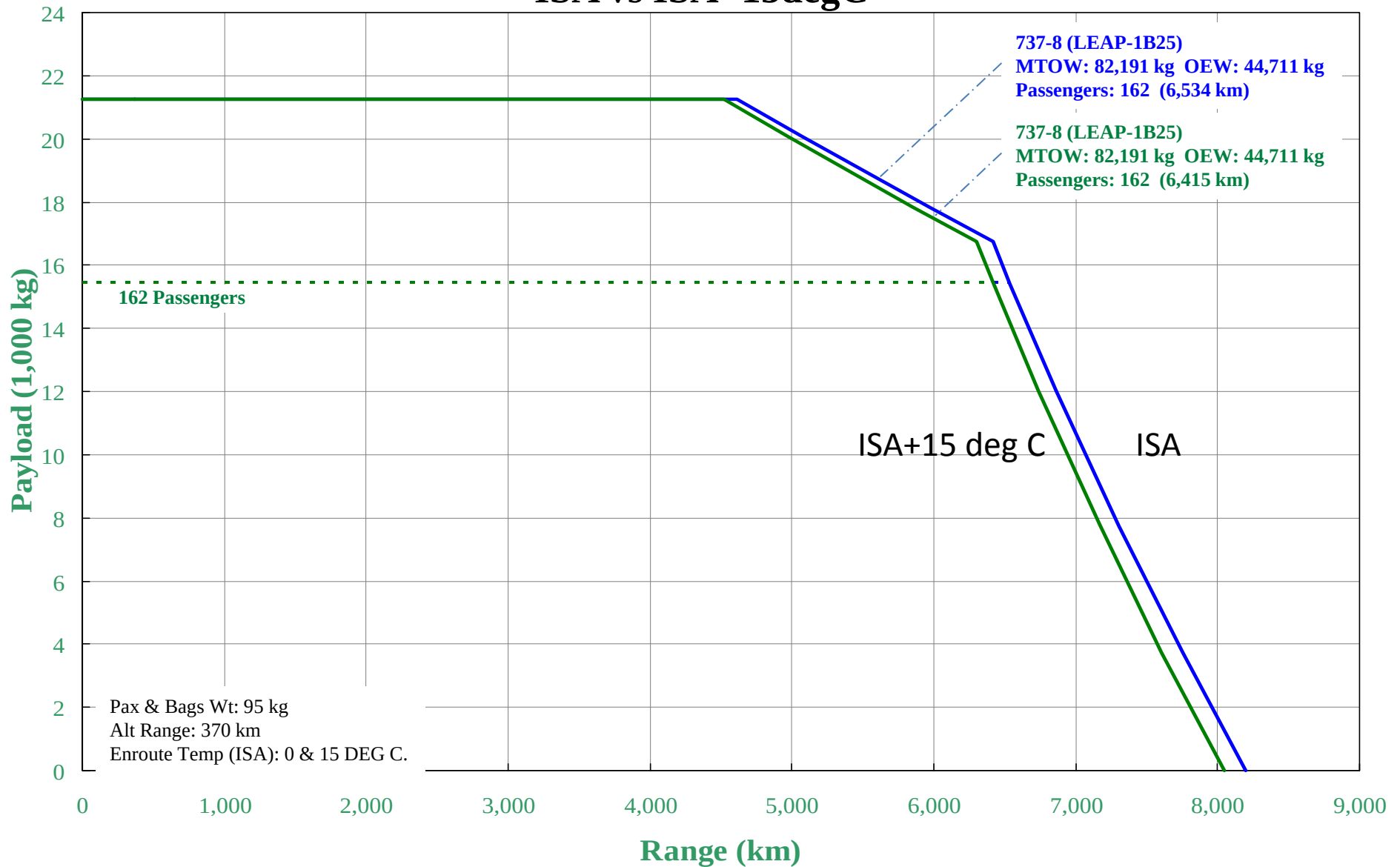
MonteCristoAir

# Payload - Range Capability - Effect of Long Alternate Distance

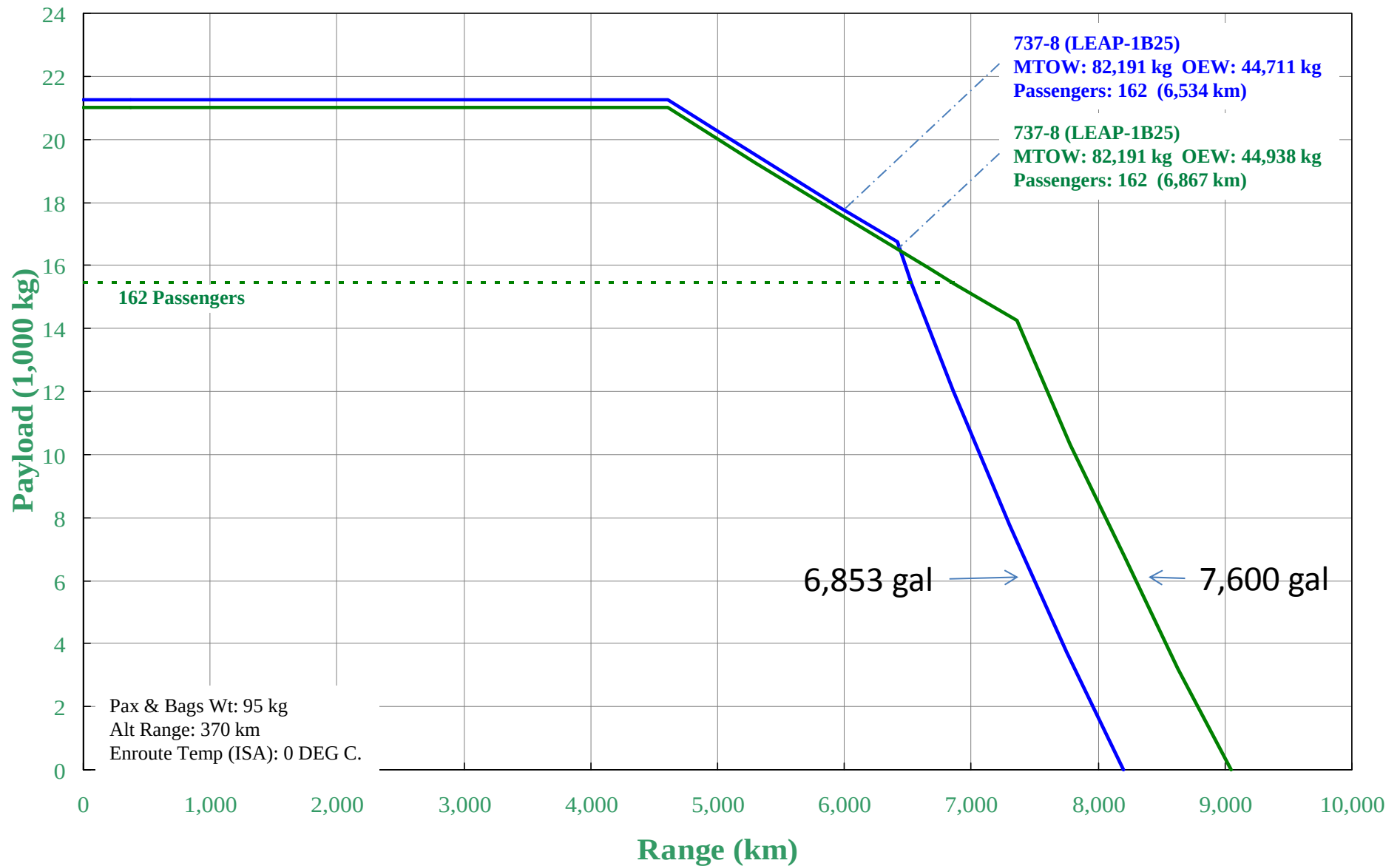
## 200nm vs 400nm



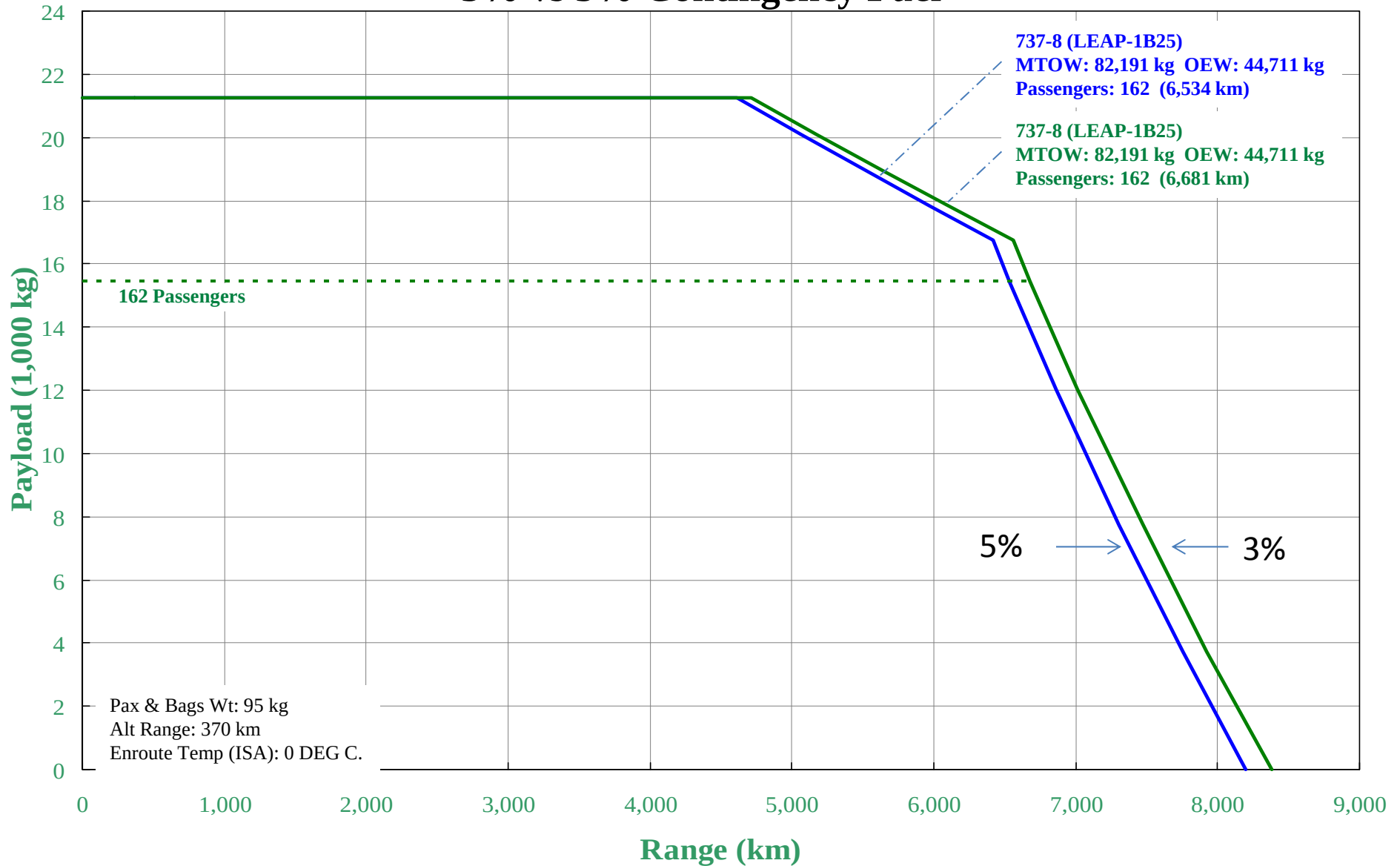
# **Payload - Range Capability - Effect of High Enroute Temperature** **ISA vs ISA+15degC**



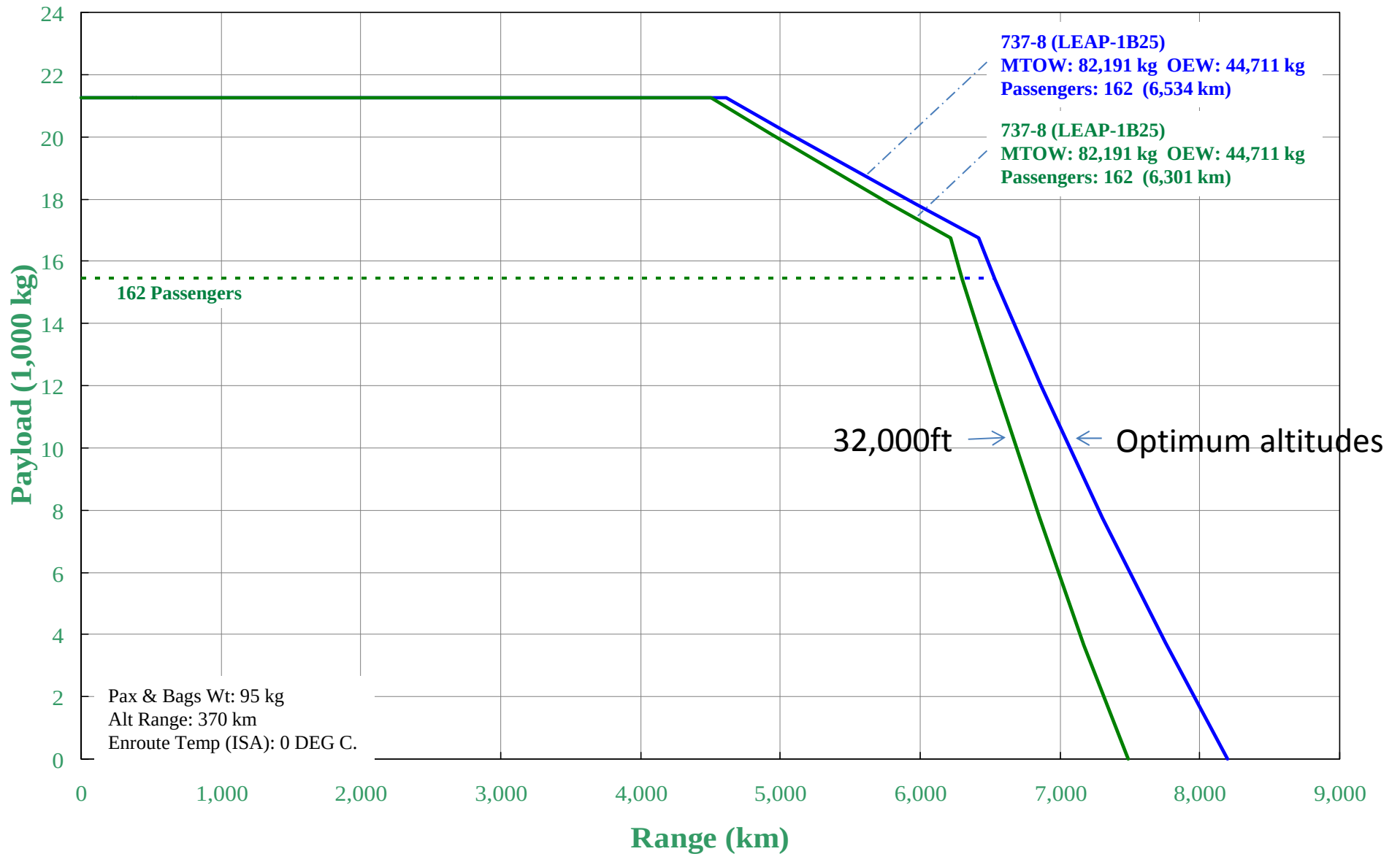
## Payload - Range Capability - Effect of Auxiliary Fuel Tank



# **Payload - Range Capability - Effect of Reduced Contingency Fuel** **3% vs 5% Contingency Fuel**



# Payload - Range Capability - Effect of Optimum vs Constant Altitude Cruise



# Payload - Range Capability - Effect of Speed on Range

## LRC vs MRC

